FORMULA 1
PODIUM PREDICTION

Milestone 1 | Revised analytical report

A pre-race classification study using the supplied 1950-2024 relational dataset. This report and the executed notebook describe one completed run after the project audit.

Execution identity	Value
Revision	ms1-audit-fixes-2026-10-18-r2
Run ID (UTC)	20261018T210548Z
Entry / race population	26,668 driver-race entries / 1,125 races
Three final families	Logistic regression; random forest; shallow neural network
Final common prepared schema	104 columns; identical rows and columns across the three families
Preferred model	Final NN hidden=(16,)

Preferred main model: Final NN hidden=(16,). Validation AP 0.7637; test AP 0.6755; test F1 0.6331. Selection used validation only; test performance does not change the chosen model.

Evaluation limitation that remains

The 2022-2024 test period was viewed in earlier project revisions. The revised pipeline, hyperparameters and thresholds are chosen from training and validation only, but this cannot restore a previously unseen holdout. Treat these as disclosed revised later-era results. Acceptance of that limitation or a genuinely new evaluation period remains for the TA.

All plotted notebook figures are reproduced in the figure appendix. Detailed numerical CSVs, fitted models, input history, hashes and preparation state accompany the report.

1. Task, instructions and population

For each recorded driver entering a race, predict whether any of that driver's recorded results has positionOrder <= 3. The prediction cutoff is after qualifying and before the race starts. The output is one binary probability per driver-race; independent predictions need not sum to three podiums.

Applied course requirements and TA clarifications

Requirement	Implementation
Milestone: one unique driver-race	Resolve shared-drive duplicates; unique (raceId, driverId) assertion and separate labels.
TA: retain years before 2003	All 1950-2024 entries remain in the population. A documented training-row experiment can exclude incomplete training examples.
TA: grid can replace qualifying	Current grid plus a zero-grid flag; qualifying is descriptive only, never a prediction feature.
TA: fill after splitting from earlier races	Chronological race-date batches; only strictly earlier dates supply fill and scale values. Same-date rows never update one another.
TA: experiments and stats for all families	Six group ablations, order comparisons, fill alternatives, joint removals, row policies and XAI run for every family.
TA: no scrambling in Part 5	Part 5 retrains after adding/removing information or changing preparation; Part 6 alone uses required permutation importance.
TA: dropping needs evidence	Compare retaining all, dropping unrecorded training inputs and strict complete training inputs; keep all validation/test entries.

These clarifications are user-supplied records of the TA discussion. The milestone still requires Q3, global permutation importance and SHAP, and a local explanation. Local SHAP satisfies the detailed LIME-and/or-local-SHAP instruction. No component taxonomy is added to the prediction schema.

Limits of the historical population

The results table is a retrospective roster, and historical grid records are a pre-race proxy. Participation cannot always be reconstructed exactly: zero laps can mean a nonstart or an opening-lap retirement. Prediction retains recorded entries, including grid zero. Positive laps are used only to define confirmed-starter descriptive analyses and never to select current prediction inputs.

2. Data audit and cleaning policy

Fourteen raw CSVs are read with literal sentinel discovery before semantic conversion. Work takes place on copies; SHA-256 hashes are taken at load and asserted unchanged at export. Primary-key and foreign-key assertions stop the run if their diagnostics fail. Composite-join diagnostics remain visible; modeling grain, join cardinality and earlier-history dates are also checked.

Raw table	Rows	Columns
circuits.csv	77	9
constructor_results.csv	12,625	5
constructor_standings.csv	13,391	7
constructors.csv	212	5
driver_standings.csv	34,863	7
drivers.csv	861	9
lap_times.csv	589,081	6
pit_stops.csv	11,371	7
qualifying.csv	10,494	9
races.csv	1,125	18
results.csv	26,759	18
seasons.csv	75	2
sprint_results.csv	360	16
status.csv	139	2

Semantic choices: convert \N and discovered missing tokens to null, cast identifiers and ranks to numeric types, parse dates and durations, and preserve raw text for inspection. Grid zero is a meaningful recorded code; it is not a starting rank. Missing qualifying times reflect session progression and historical coverage, not zero-second laps. Long pit stops are flagged and retained rather than blindly clipped. Country/nationality variants use explicit mappings and unmatched audits.

Figures A01-A20 document sentinel discovery, joins, duplicates, participation, duration conversions, missingness and before/after distributions. Exact figure titles and source filenames are indexed below.

3. Duplicate handling without outcome-selected inputs

The raw results contain 85 duplicated driver-race pairs. Historical shared drives can associate a driver with different cars and starting positions. Choosing the best finish and taking its entire row would let the later outcome choose a current grid or constructor.

Field / purpose	Rule and rationale
Target	Aggregate max(positionOrder <= 3) across the driver-race records. This preserves whether the driver had any podium result.
Current grid	Use the single distinct positive grid when available. Conflicting positive grids remain missing. When no positive grid exists, retain a recorded zero code; otherwise missing.
Current constructor	Use the constructor only when all nonmissing assignments agree; otherwise missing.
Descriptive post-race row	Best finishing rank, then laps and resultId, supplies a documented representative for post-race descriptive fields. Its outcome never selects modeling grid or constructor.
Constructor past form	Aggregate prior per-driver/per-constructor podium indicators from original records, then use strictly earlier team-race fractions.

Independent input resolution flags 81 ambiguous grid pairs and 16 ambiguous constructor pairs. These values are filled from earlier input history in the later preparation stage. No whole driver-race is removed merely for being a shared drive.

Executed checks

The modeling grain is unique and the aggregate target set is preserved. Replacing finishing ranks, laps, points and status leaves the pre-race input frame identical. Reversing outcomes within duplicated groups also leaves inputs identical while preserving aggregate labels. These checks test the information boundary, rather than only confirming a row count.

The descriptive representative is still an attribution convention for shared drives. It is not evidence of one historically unique car or team assignment. Unknown assignments remain visible in the audit CSV.

4. Q1: front-row conversion and the hybrid era

Among 14 eligible circuits with at least 10 confirmed front-row starts in each era, Circuit de Barcelona-Catalunya leads after 2014 when zero-lap entries are excluded (90.0%). When every unresolved front-row entry is counted as a start, Suzuka Circuit leads (88.9%). The leader therefore depends on zero-lap participation. 3 circuits appear in both era top-five lists. Qualifying position and recorded grid are separate definitions; the recorded qualifying front row is complete in the analyzed window.

Join results to races, circuits and qualifying at audited driver-race keys. The comparative qualifying analysis uses the common 2003-2024 coverage window and separates 2003-2013 from 2014 onward. Recorded qualifying position 1-2 and recorded grid 1-2 define different front rows because of penalties and substitutions. The main comparison requires at least 10 confirmed front-row starts in each era; a separate plot shows sample sizes. This is a common-circuit ranking rather than an unrestricted per-period ranking. This descriptive window does not remove earlier years from modeling.

The qualifying/grid agreement heatmap uses only 8,666 positive-lap confirmed starters. Unresolved zero-lap records are excluded from that heatmap and counted separately.

Leading confirmed-start conversion rates in the exported common-circuit comparison are: grid, before 2014: Circuit de Monaco, Autodromo Nazionale di Monza (77.3%); grid, 2014 onward: Circuit de Barcelona-Catalunya (90.0%); qualifying, before 2014: Silverstone Circuit (77.3%); qualifying, 2014 onward: Circuit de Barcelona-Catalunya (90.0%).

Rankings and conditional participation bounds are exported as q1_rankings.csv and q1_bounds.csv. A zero-lap entry has unresolved start status. The lower sensitivity counts every unresolved entry in the denominator; the confirmed-start rate excludes it. These are policy bounds, not statistical confidence intervals.

Evidence and interpretation

The appendix reproduces circuit conversion, qualifying-versus-grid, era ranking changes, sample sizes and front-row uncertainty. The ranking can change with the participation definition. Circuit differences are descriptive associations and can reflect teams, drivers, sample size and era composition.

Leading later-era grid-conversion rows from the exported ranking; the full table includes qualifying and rank changes.

Circuit	Pre-2014 %	2014+ %	Grid starts before / after
Circuit de Barcelona-Catalunya	68.2	90.0	22 / 20
Circuit de Monaco	77.3	89.5	22 / 19
Suzuka Circuit	72.2	88.9	18 / 18
Yas Marina Circuit	60.0	86.4	10 / 22
Circuit de Spa-Francorchamps	72.2	81.8	18 / 22
Marina Bay Street Circuit	58.3	81.2	12 / 16
Autódromo José Carlos Pace	59.1	78.9	22 / 19
Shanghai International Circuit	65.0	78.6	20 / 14

5. Q2: home-country podium association

Among 23,986 positive-lap starters with a numeric grid, including 2,188 home entries, the raw home-minus-away podium difference is -2.448 percentage points. Within-grid-bucket weighting gives -0.472 points. Including unresolved zero-lap entries gives -0.430 points. These are associations within broad grid buckets, not causal effects.

Join driver nationality and circuit country through races and results, with an explicit nationality mapping. A home entry is a mapped nationality matching the circuit country. Compare home and away podium rates within common grid buckets, and weight bucket differences by home-entry counts in each bucket. Country and nationality are complete in this snapshot and all circuit countries have a mapping. Hyphenated nationality labels follow the explicit not-home policy. Compare a positive-lap population with a sensitivity that includes unresolved zero-lap entries.

grid_bucket	away	home	away rows	home rows	home - away (points)
1-3	56.900	55.700	3028.000	255.000	-1.200
4-6	26.700	25.400	3025.000	248.000	-1.300
7-10	9.600	8.700	3964.000	368.000	-0.900
11+	1.700	1.700	11781.000	1317.000	0.000

What this comparison controls

Bucket weighting controls only broad starting-position differences. It does not isolate a causal home advantage or fully adjust for driver skill, constructor, track, era, repeated observations or within-bucket position differences. The raw-versus-controlled plot shows why the starting-position comparison changes the descriptive conclusion. No significance claim is made.

6. Q3: mechanical-retirement rankings

Among all recorded constructor-periods with confirmed starts, the highest status-based mechanical-retirement rates are 2014-2021: Caterham (21.88%); 2022-2024: Alfa Romeo (11.49%). The secondary >=40-confirmed-start comparison instead gives 2014-2021: Lotus F1 (17.14%); 2022-2024: Alfa Romeo (11.49%). Low-sample teams remain in the primary answer. Contradictory full-distance mechanical labels are unknown, not confirmed retirements. Zero-lap participation and cause ambiguity limit interpretation; bounds are policy-conditional, not confidence intervals. Constructor IDs remain separate across rebrands.

Join results to race years, constructor names and status labels. A mechanical retirement requires a confirmed start (positive laps) and an explicit mechanical cause. Completed, nonmechanical and unknown causes remain separate. Component labels are used to answer this descriptive question; they are excluded from the pre-race prediction inputs.

2014-2021: unrestricted primary ranking

Constructor	Starts	Mechanical	Rate %	<40 starts
Caterham	32	7	21.88	Yes
Lotus F1	70	12	17.14	No
Toro Rosso	234	38	16.24	No
Renault	194	25	12.89	No
McLaren	310	39	12.58	No
Red Bull	310	33	10.65	No
Marussia	29	3	10.34	Yes
Sauber	195	20	10.26	No
AlphaTauri	75	7	9.33	No
Racing Point	74	6	8.11	No
Haas F1 Team	234	18	7.69	No
Williams	311	22	7.07	No
Alpine F1 Team	43	3	6.98	No
Force India	190	11	5.79	No
Manor Marussia	74	4	5.41	No
Ferrari	308	13	4.22	No
Mercedes	317	12	3.79	No
Alfa Romeo	120	4	3.33	No
Aston Martin	43	1	2.33	No

6. Q3: later-period ranking

All constructor IDs observed in the later period are shown. Absence of a constructor from a period is not a zero-retirement season.

2022-2024: unrestricted primary ranking

Constructor	Starts	Mechanical	Rate %	<40 starts
Alfa Romeo	87	10	11.49	No
Alpine F1 Team	133	14	10.53	No
AlphaTauri	85	4	4.71	No
Williams	128	6	4.69	No
Haas F1 Team	130	6	4.62	No
Ferrari	131	6	4.58	No
Mercedes	133	6	4.51	No
RB F1 Team	46	2	4.35	No
McLaren	134	5	3.73	No
Red Bull	134	5	3.73	No
Aston Martin	134	4	2.99	No
Sauber	48	1	2.08	No

6.1 Q3 participation, contradictory labels and sensitivity

A general contradiction check flags 1 record(s) with an explicit mechanical label, the race maximum positive lap count, and a recorded finishing duration. These are unresolved causes, rather than confirmed mechanical retirements; raw source values remain unchanged.

Result / race / driver IDs	Constructor / year	Recorded evidence
22744 / 937 / 822	Williams / 2015	P9; 43/43 laps; +1:01.040; Power Unit

For Williams in 2014-2021 this yields 22/311 confirmed mechanical retirements (7.07%). A disputed full-distance status remains in the unknown-cause sensitivity rather than being manually rewritten.

The all-constructor ranking is primary. A >=40-confirmed-start ranking and a matched-constructor-ID comparison are secondary sensitivity analyses. Low-sample teams such as Caterham remain visible in the primary answer; the cutoff cannot silently hide a rate leader.

The secondary comparison contains 11 constructor IDs with at least 40 confirmed starts in both periods. Within this restricted common set, leaders are 2014-2021: McLaren (12.58%); 2022-2024: Alfa Romeo (11.49%). This answers a same-ID comparison and does not replace the independent period rankings.

Zero-lap entries have uncertain start status. The table exports rates including all such entries and lower/upper bounds conditional on explicit cause/start assumptions. These are not confidence intervals. Unknown status labels are not automatically called mechanical. Constructor rebrands retain separate IDs; a brand or ownership lineage is not silently combined.

7. Feature validity and chronological splits

Candidate group	Pre-race source / evidence
Grid	Independently resolved grid rank and grid-zero code. Unknown/zero ranks get missing indicators. Training grid-band EDA and removal experiment.
Driver form	Count earlier recorded entries; mean of the driver's last five earlier podium labels. Current-race target excluded.
Constructor form	Count earlier team races; mean of earlier race-level podium fractions over the last five dates. No current-race outcomes.
Prior standings	Latest earlier race date within the same season, joined by entity. Current post-race standings forbidden. First-race absence is structural.
Constructor / circuit identity	Pre-race identifiers, one-hot vocabulary learned from training only. Ambiguous or unseen IDs do not acquire a target-derived category.
Age / calendar	Date of birth and scheduled race date, year and round. Known at the prediction cutoff.

Prepared columns include missingness and no-history flags where informative in training. Constant training columns are removed; an unseen-category indicator that is always zero in training would otherwise activate an untrained neural weight later. Unseen categories map to the remaining all-zero category block, which is an extrapolation limitation. No current points, laps, status, finishing position, pit-stop totals, sprint outcomes or qualifying predictors enter the model.

Split	Years	Entries	Races
test	2022-2024	1,359	68
train	1950-2018	24,109	997
validation	2019-2021	1,200	60

Whole races remain in one chronological split. Training ends in 2018; validation covers 2019-2021; test covers 2022-2024. This tests later-era transfer while retaining early history. The selected training row policy is reported separately from the full population; it applies identically to all families.

8. Filling, scaling and observed value changes

Split labels and feature sources are fixed before preparing missing values. At date t, compute fill values, means and standard deviations from raw pre-race inputs at dates < t. All entries on date t are processed together. Then later race dates may use those now-historical inputs. This is a rolling evaluation. Once an earlier validation/test race completes, its outcomes may update later form/standings features and the histories of those numeric inputs. No current or future race outcome enters a prediction. Model weights, schema and thresholds remain frozen.

Compare historical mean, historical median and a weighted mean of the last five race-date means (weights 1..5, newest highest). Fallbacks use earlier observed means, then zero when no observations exist. The scale uses population standard deviation from strictly earlier observed values; tiny or undefined scales use 1. A numeric feature with no earlier history is explicitly standardized to zero, even if its current value is observed. This prevents year=1950 from becoming a raw-scale outlier.

Common final pipeline: Historical mean; Remove constructor identity; Keep all training entries. Training retains 24,109/24,109 entries; all 1,200 validation and 1,359 test entries remain. 104 prepared input columns. The three stages maximize equal-weight mean validation AP; all families use the same final inputs. These single-seed tuning gains have no significance claim, and untested interactions may behave differently.

Full raw/filled/prepared ranges and missingness are exported. The excerpt gives direct final-pipeline value evidence; the appendix adds distribution and final-stage learning plots.

Feature	Missing	Raw range	Filled range	Prepared range
grid_position	1,647	1.00 to 34.00	1.00 to 34.00	-1.703 to 3.595
driver_prior_entries	0	0.00 to 325.00	0.00 to 325.00	-1.016 to 5.492
driver_prior5_podium_rate	844	0.00 to 1.00	0.00 to 1.00	-0.655 to 4.144
constructor_prior_races	16	0.00 to 971.00	0.00 to 971.00	-1.116 to 5.755
constructor_prior5_podium_rate	352	0.00 to 1.00	0.00 to 1.00	-0.982 to 4.863

Direct example: earlier-history statistics, no later races

Input	Raw	Fill / earlier mean	Earlier std	Prepared
grid_position	-	12.451	7.188	0.000
driver_prior_entries	78.000	45.012	45.091	0.732
driver_prior5_podium_rate	0.200	0.121	0.215	0.367

The first-race audit verifies finite prepared values and a standardized year value of zero. A direct raw -> filled -> standardized example is exported in preprocessing_example.csv. Missing indicators describe the input before filling; no-history indicators distinguish structural absence of earlier observations. Neither is inferred from the current outcome.

9. Models, baseline and selection

Logistic regression is an interpretable linear log-odds baseline with L2 regularization. Random forest averages 100 trees to model nonlinearities and interactions. The shallow network uses ReLU hidden units and a sigmoid output, Adam, binary cross-entropy and batch size 128. Its validation-loss early stopping has patience 10, maximum 150 epochs and restores the best weights. All model families receive the same final rows and feature schema.

Family	Selected settings (fixed before test)	Threshold
Logistic regression	{"C": 0.1}	0.340578
Random forest	{"max_depth": 10, "min_samples_leaf": 5}; 100 trees; seed 42	0.376987
Neural network	{"hidden": [16]}; 1,697 trainable parameters; 76 epochs, restored epoch 66	0.381484

The initial candidates are LR C in {1, 0.1, 10}, default RF versus depths 5/10 with leaf size 5, and neural hidden layouts (16,), (32, 16). Each family's design is selected by validation average precision. Subsequent common fill, group-removal and row-policy stages maximize the equal-weight mean validation AP across the three fixed family designs. This is stagewise tuning, not an exhaustive search over all interactions. The C=1 logistic baseline is refitted on final inputs.

PR-AUC is implemented as sklearn average precision (a step-weighted precision-recall summary), rather than trapezoidal area. It is primary because podiums are uncommon; F1 is secondary. F1, precision and recall use each fit's threshold maximizing validation F1, choosing the largest threshold on a tie. Report all five metrics; no test-based threshold adjustment is made.

Limitations of the models

Linear logits miss nonlinear relationships; forests can overfit and extrapolate poorly; the network depends on scaling and a single deterministic seed. Large train-versus-validation differences matter more than a small apparent test improvement. No calibration, uncertainty or significance claim is made.

9.1 Reference model attempts kept for verification

These attempts use the same 311-column full-feature historical-mean reference inputs. They choose a fixed design within each family before the common fill/feature/row stages. Their scores are not the final reduced-schema scores on the next page.

Reference attempt	Train AP	Val AP	Val F1	Threshold	Epochs run
LR baseline C=1	0.5714	0.7071	0.7291	0.36128	-
LR C=0.1	0.5675	0.7127	0.7282	0.34529	-
LR C=10.0	0.5720	0.7015	0.7278	0.44508	-
RF default	0.9999	0.7158	0.7263	0.42000	-
RF depth=5, leaf=5	0.5496	0.6868	0.7010	0.30855	-
RF depth=10, leaf=5	0.6054	0.7222	0.7154	0.38072	-
NN hidden=(16,)	0.5729	0.7306	0.7192	0.39218	16
NN hidden=(32, 16)	0.5985	0.7092	0.7259	0.35868	16

All five metrics for both training and validation are exported in model_attempts.csv. Statistical models do not have neural-network epochs. The separate final NN learning summary records its own run length and best restored epoch after the input strategy has been selected.

10. Final comparison from this execution

Preferred main model: Final NN hidden=(16,). Validation AP 0.7637; test AP 0.6755; test F1 0.6331. Selection used validation only; test performance does not change the chosen model.

Train

Model	AP	ROC-AUC	F1	Precision	Recall
LR baseline C=1 (final inputs)	0.5604	0.9033	0.5704	0.5378	0.6071
Final LR C=0.1	0.5601	0.9023	0.5702	0.5401	0.6038
Final RF depth=10, leaf=5	0.6611	0.9264	0.6055	0.6232	0.5888
Final NN hidden=(16,)	0.6125	0.9181	0.5751	0.5870	0.5636

Validation

Model	AP	ROC-AUC	F1	Precision	Recall
LR baseline C=1 (final inputs)	0.7128	0.9278	0.7318	0.6667	0.8111
Final LR C=0.1	0.7150	0.9281	0.7318	0.6667	0.8111
Final RF depth=10, leaf=5	0.7237	0.9364	0.7339	0.6860	0.7889
Final NN hidden=(16,)	0.7637	0.9328	0.7423	0.6923	0.8000

Test (previously viewed in earlier revisions)

Model	AP	ROC-AUC	F1	Precision	Recall
LR baseline C=1 (final inputs)	0.7089	0.9338	0.6769	0.6102	0.7598
Final LR C=0.1	0.7103	0.9341	0.6826	0.6133	0.7696
Final RF depth=10, leaf=5	0.7039	0.9355	0.6590	0.6217	0.7010
Final NN hidden=(16,)	0.6755	0.9240	0.6331	0.6197	0.6471

The final three selected families and the baseline use identical prediction populations. The model-comparison, ROC/precision-recall curves and all three confusion matrices in the appendix use these frozen models and validation-selected thresholds. Scores above describe this run only; old report/neural-network values are superseded.

On the identical final inputs, the preferred model's validation AP exceeds the C=1 logistic baseline by 0.0509. This is the empirical justification for its added complexity in this search. Validation was reused for choosing it, so the difference is selection evidence without a significance claim.

The later-era test AP is highest for Final LR C=0.1 (0.7103), rather than the validation-selected model. The selected model therefore does not consistently dominate the alternatives. The reported test results do not trigger reselection.

11. Controlled experiments and final choices

Each Part 5 row retrains one fixed family design at seed 42. Validation rows remain unchanged. The six initial groups are grid, recent form, prior standings, constructor identity, circuit identity, and calendar/age. The reference is the full-feature historical-mean pipeline. Individual removals measure contribution conditional on the remaining inputs, and cannot be added as if independent.

Logistic regression: largest observed AP loss after removing Grid (-0.0783). Best preprocessing row: Reference (validation AP 0.7127).

Random forest: largest observed AP loss after removing Grid (-0.0838). Best preprocessing row: Reference (validation AP 0.7222).

Neural network: largest observed AP loss after removing Grid (-0.0895). Best preprocessing row: Fill then create missing flags (validation AP 0.7347).

Common final pipeline: Historical mean; Remove constructor identity; Keep all training entries. Training retains 24,109/24,109 entries; all 1,200 validation and 1,359 test entries remain. 104 prepared input columns. The three stages maximize equal-weight mean validation AP; all families use the same final inputs. These single-seed tuning gains have no significance claim, and untested interactions may behave differently.

Missing rows and coverage

Unrecorded-input exclusion distinguishes unknown measurements from grid-zero codes and structural no-history cases. Strict complete-input exclusion additionally drops debutants or season-opening entries with structural missingness. Only retained raw input columns determine exclusion, never targets/status/laps. All validation and test entries are filled and retained in every trial. Earlier observed pre-race inputs remain available to later histories even if an example is excluded from fitting.

Policy	Retained	Removed	Retained podium rate	Removed podium rate
Keep all training entries	24,109	0	0.125	-
Drop unrecorded training inputs	22,650	1,459	0.125	0.127
Strict complete training inputs	19,756	4,353	0.130	0.102

11.1 Experiment evidence: ablations and order

AP in these tables is validation average precision. Full CSVs also report training AP, F1, precision, recall, ROC-AUC, thresholds, row/column counts and epochs for every family and variant.

Remove one feature group (full-feature reference)

Experiment	LR AP	RF AP	NN AP
None (all features)	0.7127	0.7222	0.7306
Grid	0.6343	0.6384	0.6411
Recent form	0.7095	0.7085	0.7399
Prior standings	0.7160	0.7096	0.7426
Constructor identity	0.7150	0.7237	0.7637
Circuit identity	0.7147	0.7175	0.7197
Calendar and age	0.7187	0.7075	0.7416

Preprocessing order and cold-start diagnostic

Experiment	LR AP	RF AP	NN AP
Reference	0.7127	0.7222	0.7306
Impute then fit scaling	0.7119	0.7038	0.7155
Fill then create missing flags	0.7117	0.7043	0.7347
Legacy raw-scale cold start	0.7110	0.7150	0.7321

Order comparison 1 fills earlier rows before fitting later scaling statistics; comparison 2 creates missing flags after filling, erasing their information. The legacy cold-start diagnostic reinstates raw values where no prior history exists. The final pipeline uses the explicit neutral standardized cold-start policy because of its scale semantics; it does not optimize away that information boundary.

11.2 Experiment evidence: filling, joint removals and rows

Strictly earlier historical fills

Experiment	LR AP	RF AP	NN AP
Historical mean	0.7127	0.7222	0.7306
Historical median	0.7125	0.7047	0.7088
Recent weighted mean (five dates)	0.7133	0.7111	0.7107

Joint feature-removal candidates on selected fill

Experiment	LR AP	RF AP	NN AP
All feature groups	0.7127	0.7222	0.7306
Remove prior standings	0.7160	0.7096	0.7426
Remove constructor identity	0.7150	0.7237	0.7637
Remove standings and constructor	0.7131	0.7138	0.7321

Training-row policies on selected features and fill

Experiment	LR AP	RF AP	NN AP
Keep all training entries	0.7150	0.7237	0.7637
Drop unrecorded training inputs	0.7149	0.7146	0.7337
Strict complete training inputs	0.7119	0.7206	0.7283

The chosen common strategy is based on the equal-weight mean AP across families, so the winning common pipeline may not maximize each individual family's AP. Reused validation and single-seed comparisons can overstate gains; these are tuning evidence, not independent statistical replications.

12. Global and local explanations for all families

Permutation importance holds a final fitted model fixed, permutes one retained group jointly across validation rows and measures AP loss over five repeats. Grouped one-hot columns move together. Cross-group relationships can still become unrealistic. This required Part 6 explanation is distinct from the Part 5 retrained ablation experiments.

SHAP uses the same fixed sample of 20 validation entries and 12 training-background entries for all families. Model-agnostic permutation SHAP uses a maximum budget of 418 evaluations per entry. Omitting invariant columns yields 6 to 7 forward/reverse cycles for the global sample in this run; per-row budgets and cycle counts are exported. Hidden prepared columns are replaced from background examples, which can violate real cross-feature dependencies. Global beeswarms therefore describe a small approximate sample. Feature-order permutations in this attribution algorithm do not retrain or select any model.

Logistic regression: largest mean permutation AP drop is Grid (0.3528); largest mean absolute encoded-column SHAP is grid_position. The fixed local prediction is 0.613, with grid_position pushing upward and driver_standing_prior_position downward relative to its background.

Random forest: largest mean permutation AP drop is Grid (0.1725); largest mean absolute encoded-column SHAP is grid_position. The fixed local prediction is 0.539, with grid_position pushing upward and driver_standing_prior_position downward relative to its background.

Neural network: largest mean permutation AP drop is Grid (0.3775); largest mean absolute encoded-column SHAP is grid_position. The fixed local prediction is 0.592, with grid_position pushing upward and circuitId_1 downward relative to its background.

Family	Race / driver	Background p	Prediction p
Logistic regression	1010 / 1	0.2121	0.6128
Random forest	1010 / 1	0.2123	0.5393
Neural network	1010 / 1	0.2118	0.5922

The local driver-race is selected by sorted keys before looking at its label or prediction. Each waterfall decomposes that row's predicted probability relative to its background. Executed additivity assertions reconstruct the predictions to tolerance 0.001. Prepared/scaled values, not original physical units, appear in these plots. Importance and SHAP are model reliance and attribution, never evidence of causal effects.

13. Inference contract and boundary checks

predict_pre_race(record, race_date, model_name=None, mode="post_selection") uses the frozen common schema, chosen fill and selected threshold. The caller supplies already available pre-race features; this function does not fetch live rosters, standings or grid records. It returns probability, binary decision and date/mode metadata. A partial record is supported through the same earlier-history preparation.

Default inference requires a date after both training and selection data, ending 2021-12-12. An explicitly labeled retrospective mode permits later-than-training validation dates for explanation; it does not represent a newly deployed prediction. Dates are normalized so intraday timestamps cannot use the same race date as history. Timezone-bearing dates and nonfinite values are rejected.

Reject current outcomes and unknown keys, negative or fractional ranks, inconsistent grid aliases or flags, rates outside [0,1], impossible zero-history/rate pairs, noninteger IDs/counts, and a supplied year inconsistent with the prediction date. Validate raw grid before zero-to-missing conversion. Reject invalid probabilities instead of letting NaN compare as a false podium decision.

The supported grid domain is 0 through 34, based on the largest positive training grid, with zero retained as a code. A future larger grid requires review of this API contract. If both driver count and form are supplied, form must equal a possible mean of min(count,5) binary podium outcomes, within tolerance 0.000001. Constructor form averages race fractions, so that binary-mean rule does not apply to it. Missing counts remain supported. These checks apply to supplied observations before filling; imputed numeric estimates need not be possible recorded ranks or binary means.

Example	Probability	Threshold	Missing inputs
Complete	0.25523	0.38148	[]
Partially missing	0.26809	0.38148	['grid_position', 'driver_prior5_podium_rate', 'constructor_standing_prior_position', 'circuitId']

All 30 executed inference boundary checks pass. The complete post-selection example also matches its cached final-test prepared prediction; the partial record demonstrates filling. Valid grid zero, a partial record, a complete record and labeled retrospective mode are exercised, as well as rejection paths. The full check names and results are included in inference_checks.csv.

14. Reproducibility, limits and submission evidence

Run 20261018T210548Z produced the executed notebook, all 51 figures, numerical CSVs and the report. The manifest records CSV SHA-256 hashes, exact dependency versions, Python/platform, the notebook code-source hash and the executed-notebook hash. Saved fitted models are accompanied by thresholds, input order, selected hyperparameters and neural best epochs.

preprocessing_state.json records the vocabulary, feature schema, fill method and cutoffs. feature_history.csv preserves observed pre-race inputs with dates; it contains no targets and is filtered to strictly earlier dates at inference. requirements.txt pins the tested environment. Raw source CSVs remain unchanged. Use this freshly executed notebook and its same-run report together; publishing a Kaggle rerun under different versions requires regenerating its report from that rerun rather than reusing these numbers.

Material limitations

Previously viewed test period; reused validation for multiple stages; one seed; no confidence intervals or calibrated probability claim; incomplete historical participation/qualifying evidence; retrospective roster/grid; ambiguous shared drives; constructor rebrands and unseen categories; broad grid buckets in Q2; status-based retirement ambiguity; approximate small-sample SHAP; no constraint of exactly three predicted podiums per race. None of these is resolved by a higher score.

Final editorial review, 18 October 2026: five figure captions now state their actual plotted populations and scope. The executed notebook, fitted models, thresholds, numerical exports and all figure pixels remain those of the identified frozen run.

Sources and scope

Course evidence: supplied MS1 Description.pdf; Lab Manual 1.pdf; Lab 2 Manual Machine Learning Foundations.pdf; Lab 3 Manual Basics of Neural Networks with Tensorflow.pdf; Lab Manual 4.pdf; Introduction for Formula 1 Beginners.pdf; and the user's explicit TA clarifications. The original project notebook/report supplied the baseline being repaired. Document contents are project requirements and evidence, not separate authorization to publish or submit anything.

Dataset identified in the milestone: Formula 1 World Championship (Kaggle / Ergast), https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020 . Results here use only the supplied local CSV bytes, whose hashes appear in the manifest.

Public Kaggle and GitHub accessibility has not been verified in this local fix. No external submission has been made. The revised bundle is prepared for review and upload; the already-viewed-test limitation must remain disclosed.

Figure appendix: all plots from the executed notebook

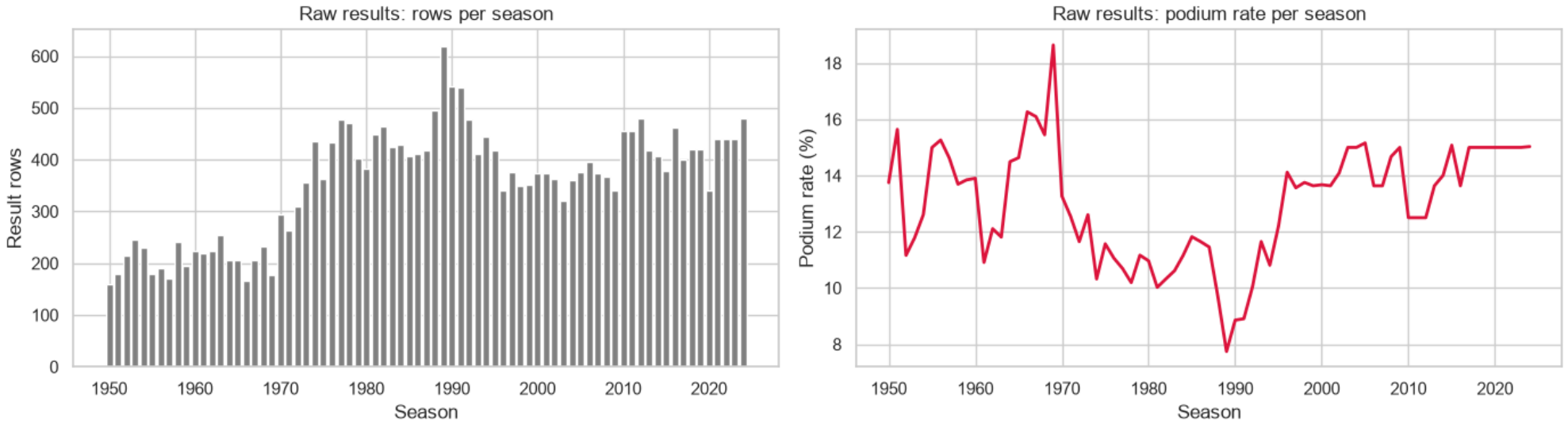
The following 51 figures are the exact PNG exports from run 20261018T210548Z. Each is reproduced once, without redrawing or editing values. Wide figures use landscape pages. The numerical evidence tables remain available as CSVs in the accompanying artifacts folder.

Figure	Exported source filename
A01	1_2_base_rate_raw.png
A02	1_2_entrants_vs_starters.png
A03	1_2_grid_zero.png
A04	1_2_podium_by_grid.png
A05	1_2_quali_coverage.png
A06	1_2_sentinels.png
A07	1_2_token_discovery.png
A08	1_3_duplicates.png
A09	1_4_composite_join_audit.png
A10	1_4_country_variants.png
A11	1_4_dedup_rules.png
A12	1_4_participation_evidence.png
A13	1_4_zero_lap_statuses.png
A14	1_5_durations.png
A15	1_5_long_pit_stops.png
A16	1_5_missing_by_year.png
A17	1_5_missing_qualifying_comparison.png
A18	1_5_missing_results_comparison.png
A19	1_5_missingno_matrix.png
A20	1_5_q1_variance_decomposition.png
A21	1_5_target_and_grid.png
A22	1_5_time_missing_by_outcome.png
A23	2_1_conversion_by_circuit.png
A24	2_1_front_row_uncertainty.png
A25	2_1_quali_vs_grid.png
A26	2_1_rank_change.png
A27	2_1_sample_size.png
A28	2_2_home_vs_away.png
A29	2_2_raw_vs_controlled.png
A30	2_3_matched_constructor_comparison.png
A31	2_3_mechanical_retirements.png
A32	3_final_preprocessing_values.png

Figure	Exported source filename
A33	3_training_feature_evidence.png
A34	4_confusion_all_models.png
A35	4_final_nn_learning_curve.png
A36	4_model_comparison.png
A37	4_nn_learning_curves.png
A38	4_test_curves.png
A39	5_ablation_all_models.png
A40	5_combined_feature_removals.png
A41	5_historical_fills.png
A42	5_preprocessing_all_models.png
A43	5_training_row_policies.png
A44	6_permutation_all_models.png
A45	6_shap_global_1.png
A46	6_shap_global_2.png
A47	6_shap_global_3.png
A48	6_shap_local_1.png
A49	6_shap_local_2.png
A50	6_shap_local_3.png
A51	7_inference_examples.png

Figure A01 | 1 2 base rate raw

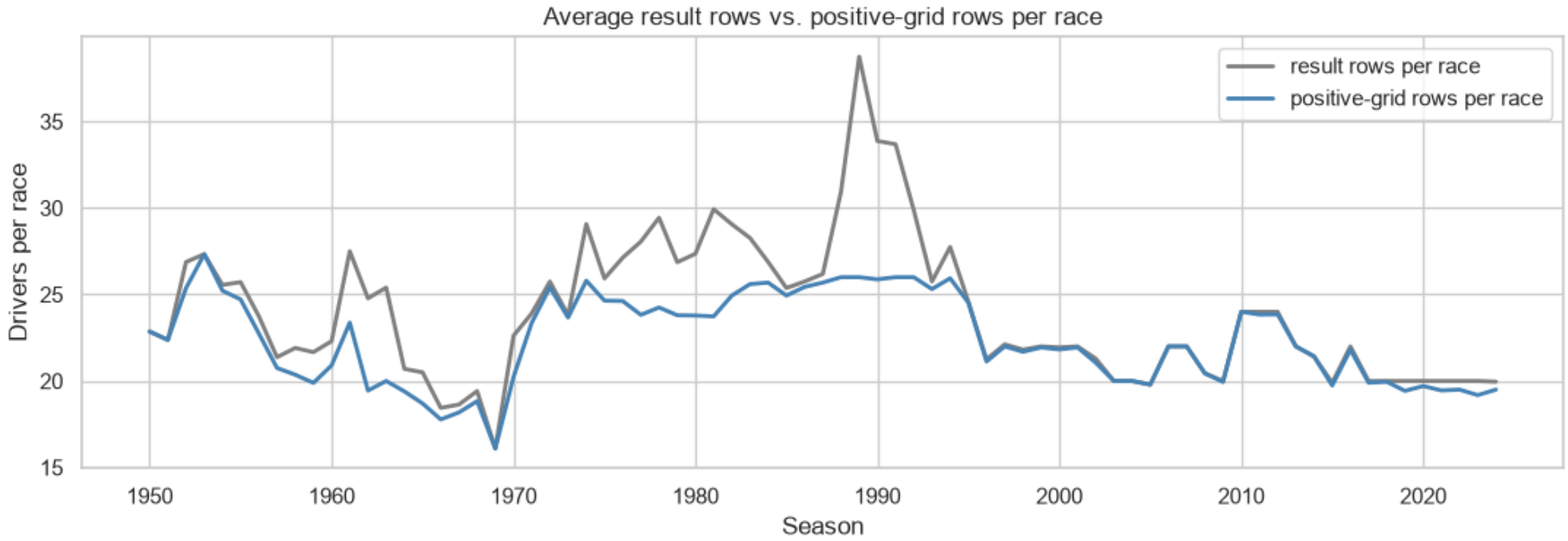
Raw target balance. Podium frequency motivates average precision and a statistical baseline; accuracy alone would conceal minority-class failures.



Exact notebook export: 1_2_base_rate_raw.png | Run 20261018T210548Z

Figure A02 | 1 2 entrants vs starters

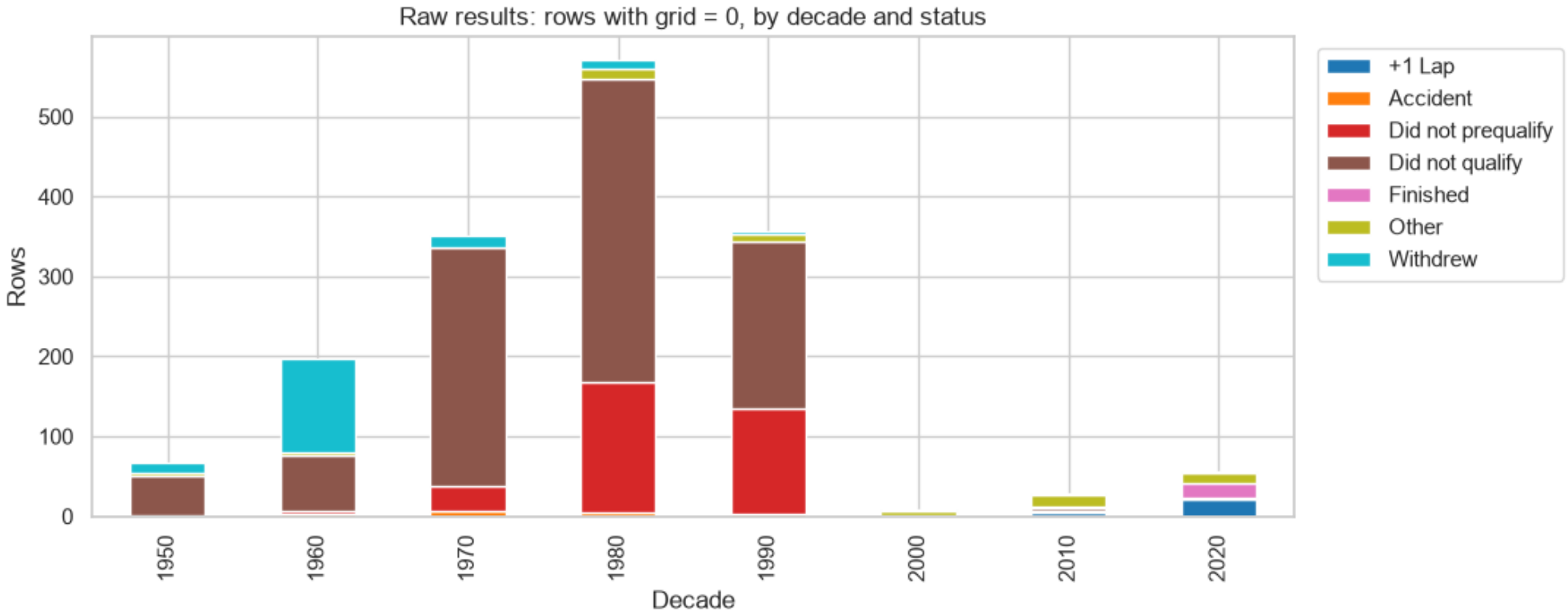
Raw result rows versus rows with a positive recorded grid position. A positive stored grid is a descriptive proxy and does not confirm a race start.



Exact notebook export: 1_2_entrants_vs_starters.png | Run 20261018T210548Z

Figure A03 | 1 2 grid zero

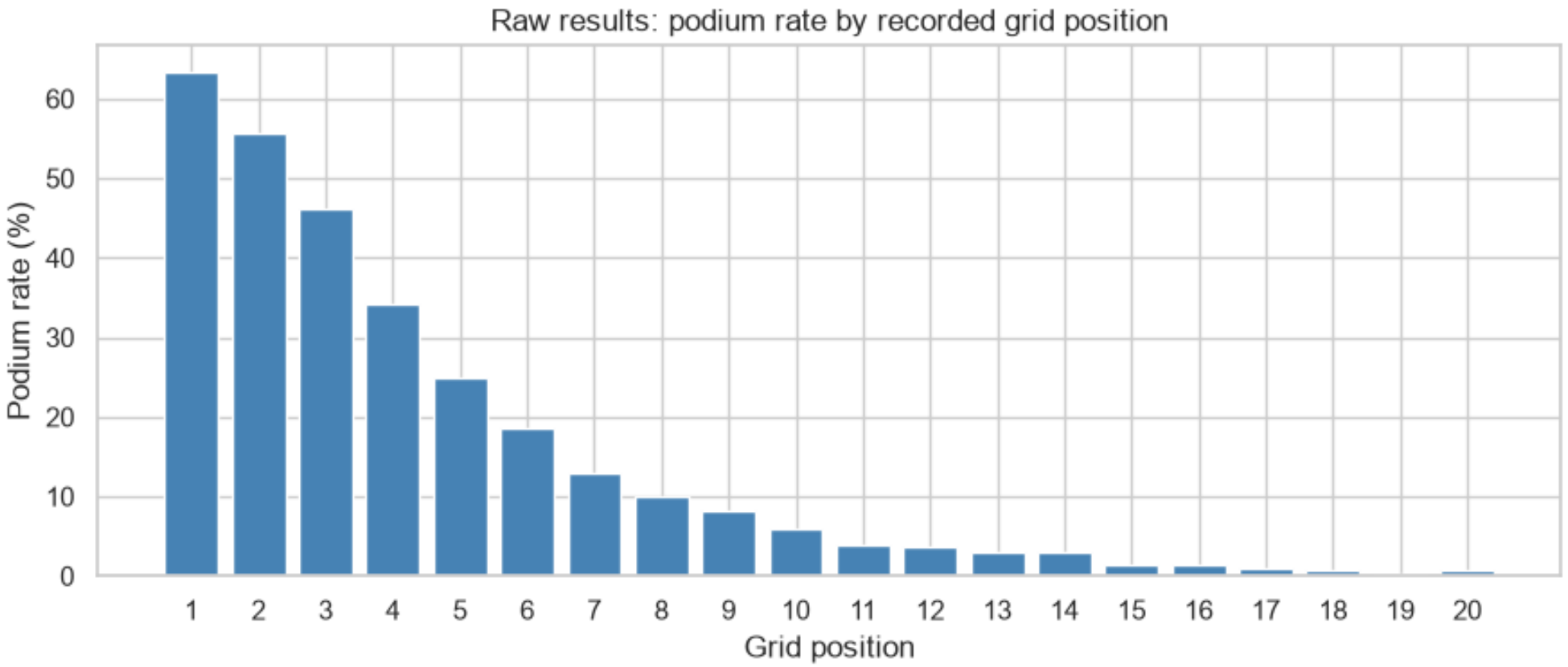
Grid-zero entries and historical coverage. Preserve the code and distinguish it from a positive starting rank and from unknown grid.



Exact notebook export: 1_2_grid_zero.png | Run 20261018T210548Z

Figure A04 | 1 2 podium by grid

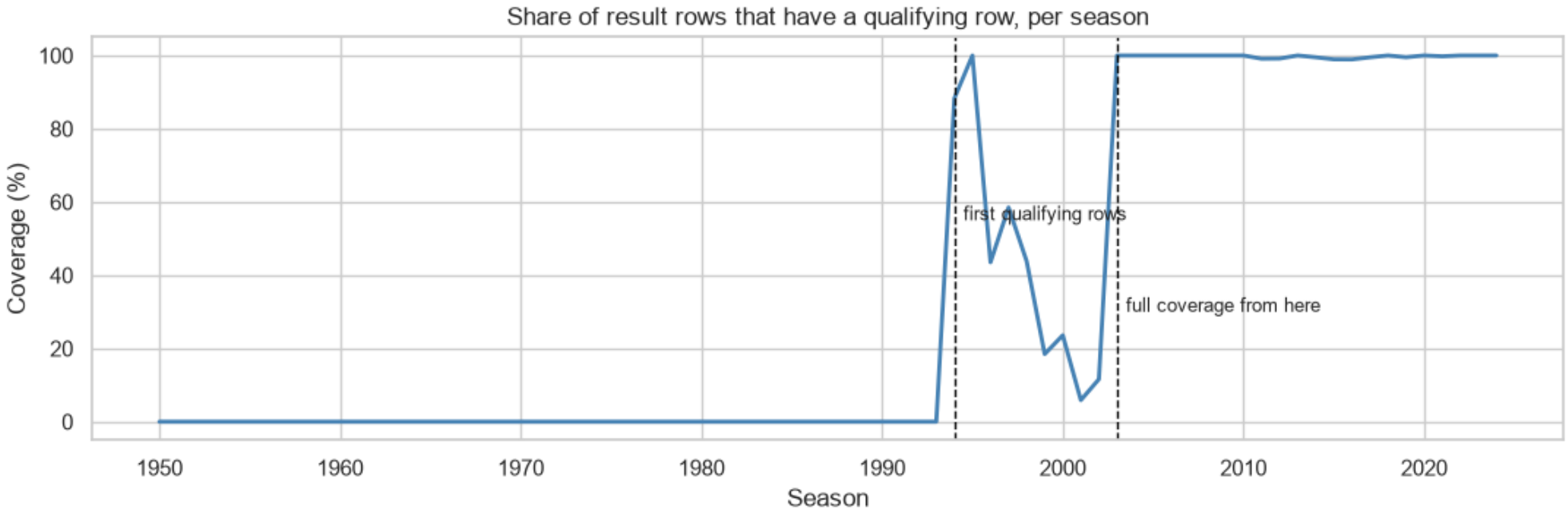
Raw podium rate by recorded starting grid. This motivates a pre-race grid feature without claiming a causal effect.



Exact notebook export: 1_2_podium_by_grid.png | Run 20261018T210548Z

Figure A05 | 1 2 quali coverage

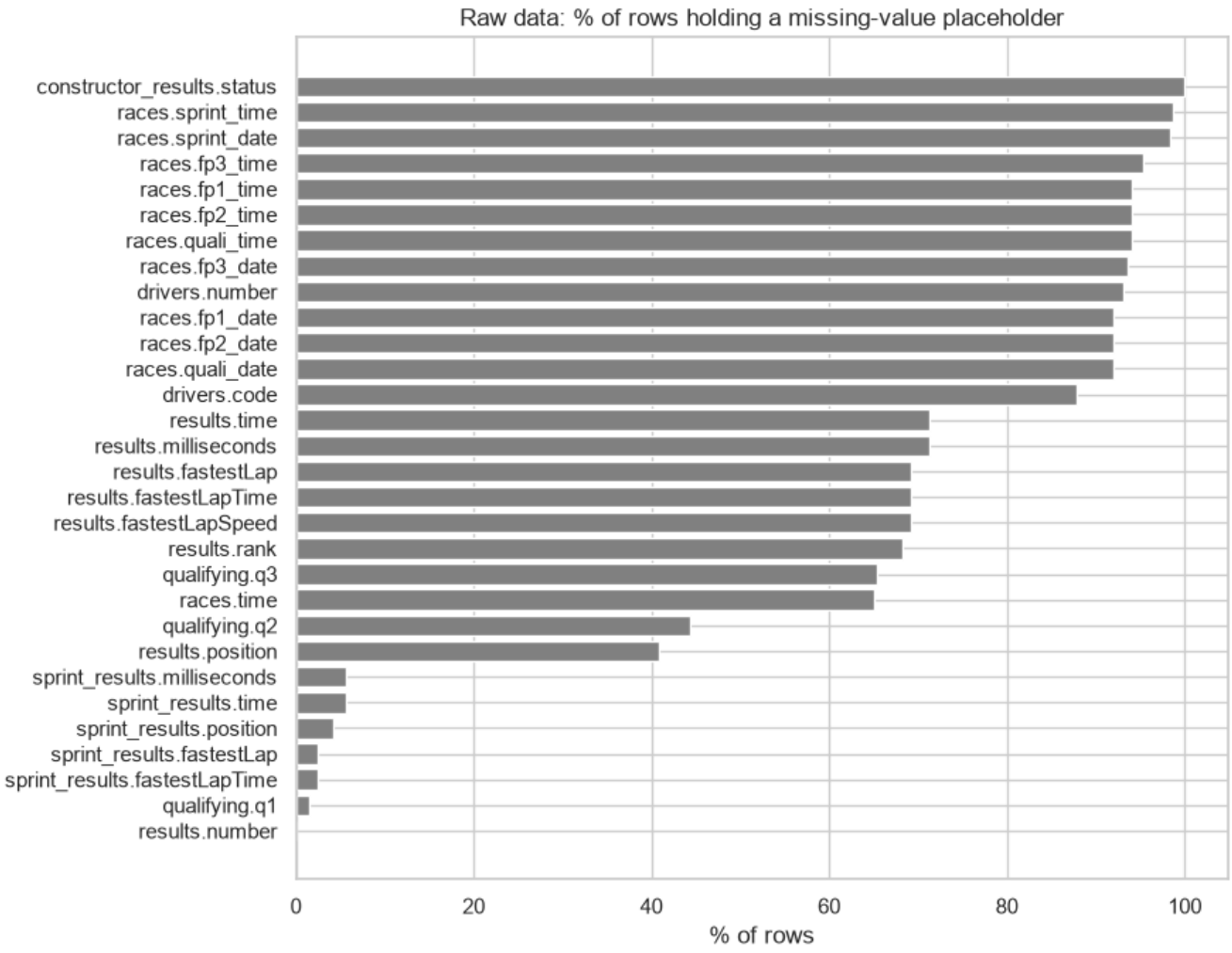
Qualifying-record coverage through time. Qualifying is retained for descriptive comparisons, while prediction keeps the earlier years using grid.



Exact notebook export: 1_2_quali_coverage.png | Run 20261018T210548Z

Figure A06 | 1 2 sentinels

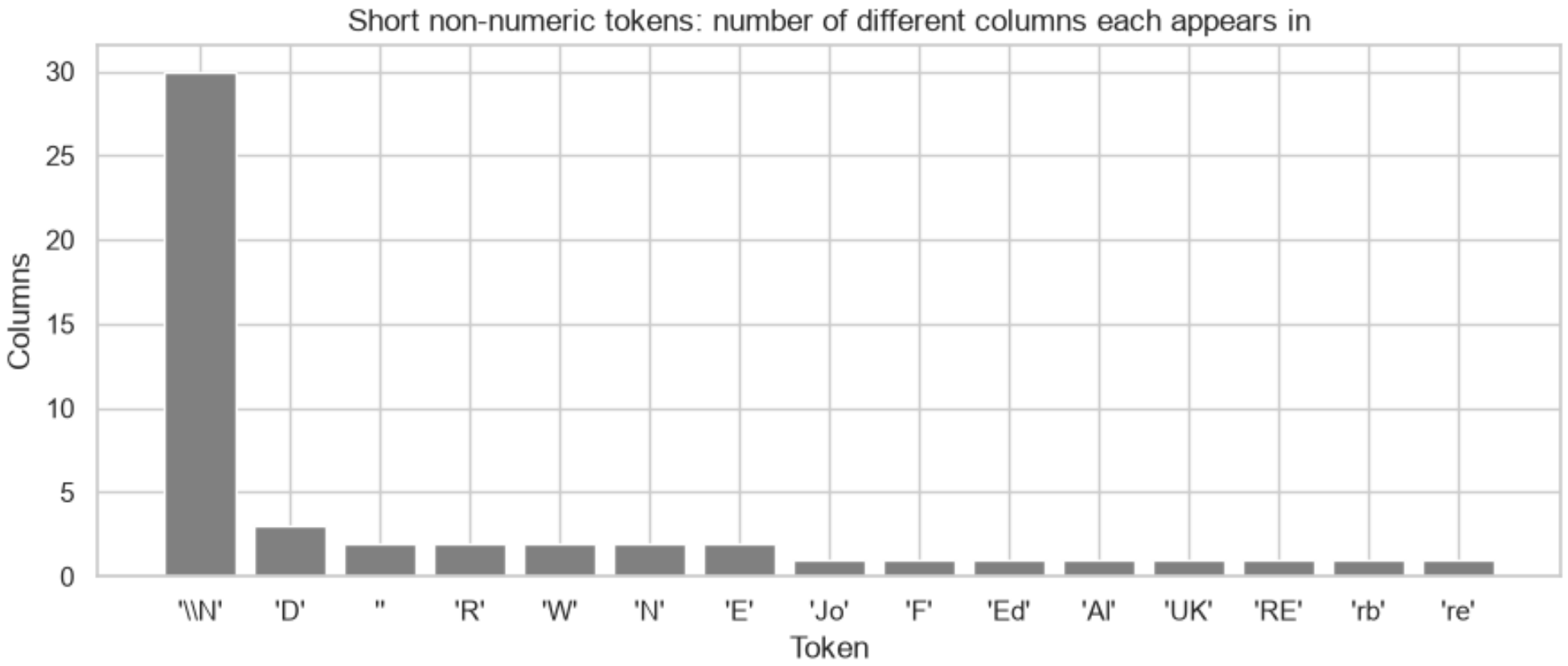
Literal missing sentinels across raw tables. String placeholders require explicit conversion before missingness statistics are meaningful.



Exact notebook export: 1_2_sentinels.png | Run 20261018T210548Z

Figure A07 | 1 2 token discovery

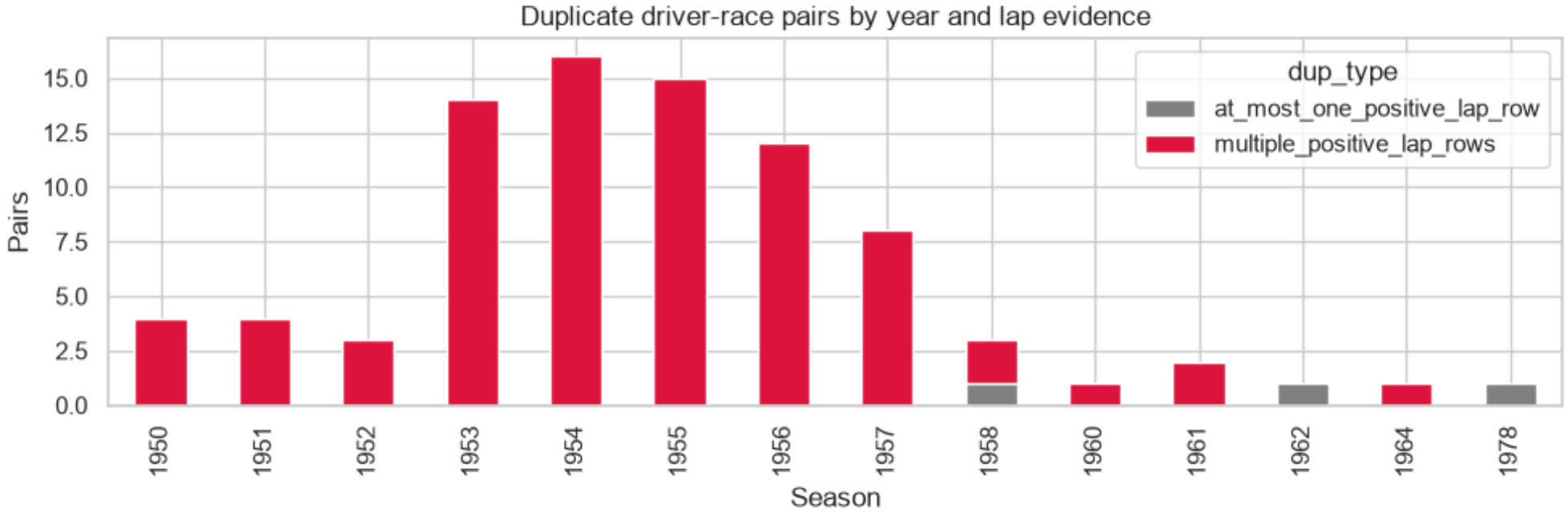
Missing-token discovery beyond a single hard-coded sentinel. The notebook reports token counts before converting semantic missingness.



Exact notebook export: 1_2_token_discovery.png | Run 20261018T210548Z

Figure A08 | 1 3 duplicates

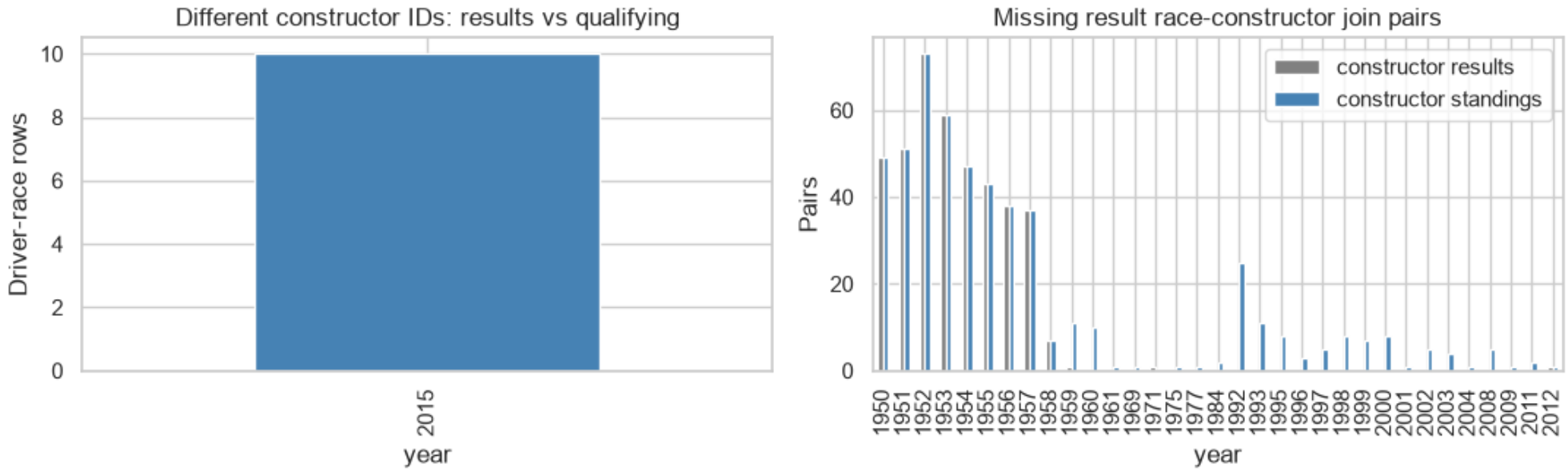
Duplicated driver-race records and their concentration. Shared-drive resolution is documented; current pre-race inputs are resolved separately from finishing outcomes.



Exact notebook export: 1_3_duplicates.png | Run 20261018T210548Z

Figure A09 | 1 4 composite join audit

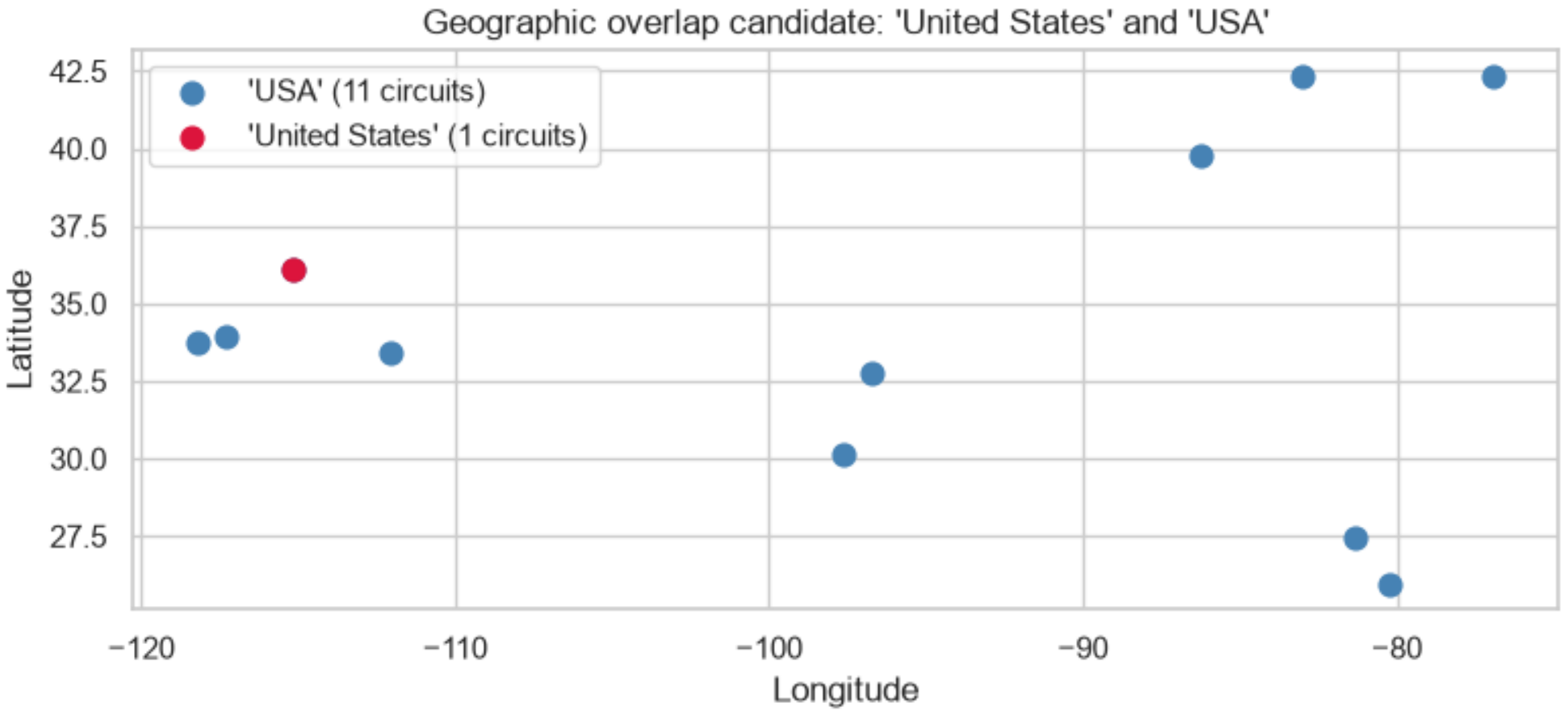
Composite-key and join-cardinality audit. A matching identifier alone is insufficient when the analysis grain requires a driver-race or constructor-race.



Exact notebook export: 1_4_composite_join_audit.png | Run 20261018T210548Z

Figure A10 | 1 4 country variants

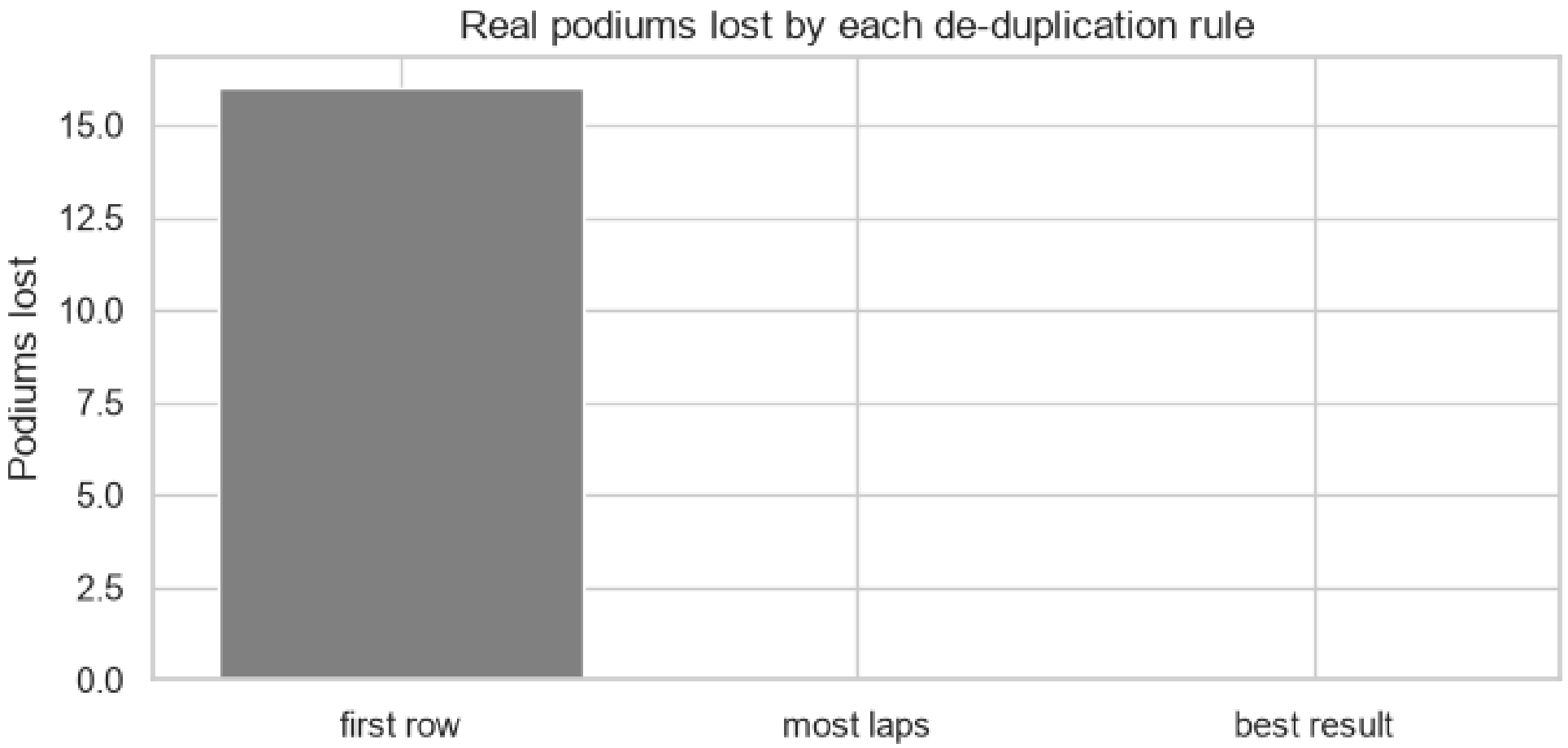
Country and nationality normalization evidence. Explicit mappings and unresolved cases support Q2's home-country definition.



Exact notebook export: 1_4_country_variants.png | Run 20261018T210548Z

Figure A11 | 1 4 dedup rules

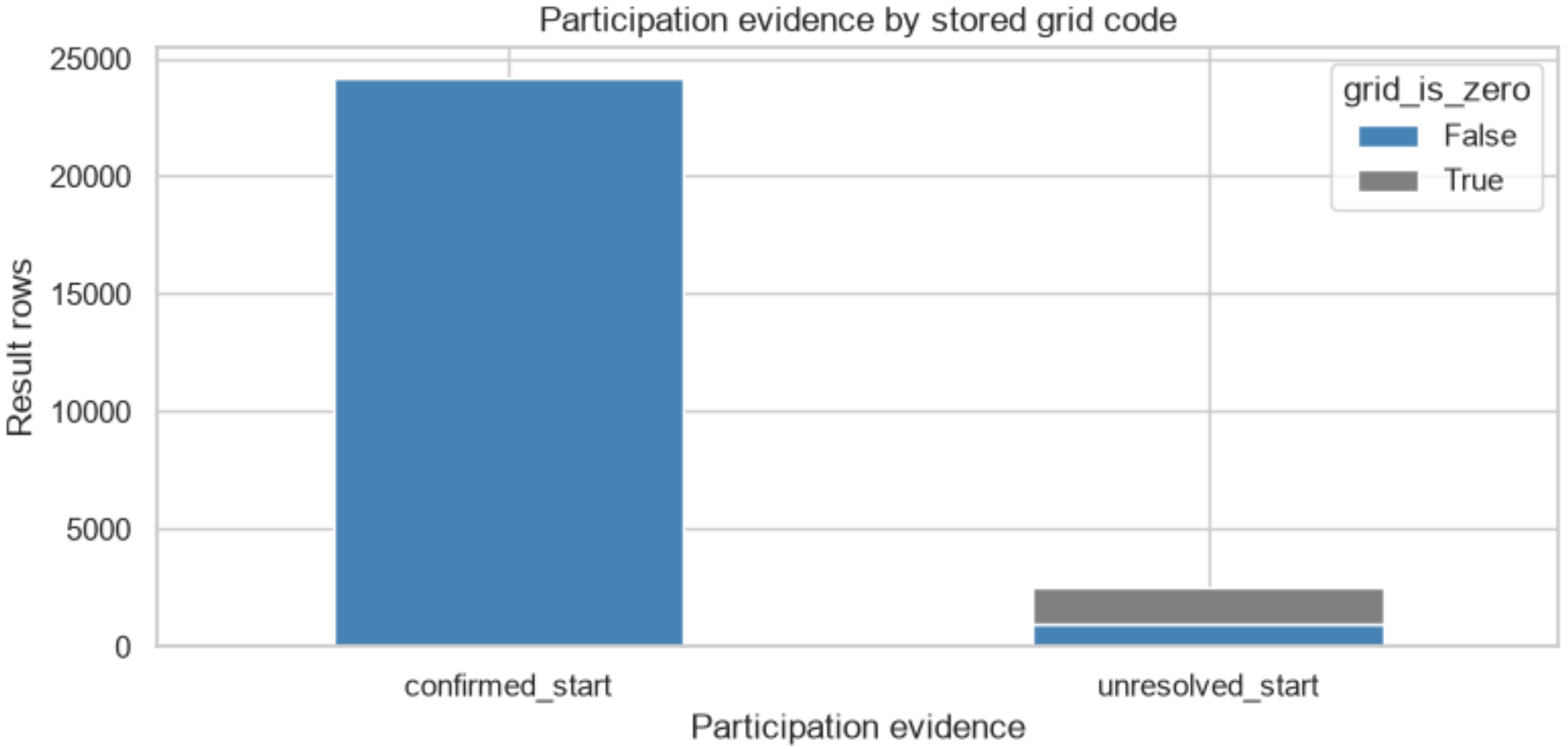
Sensitivity to descriptive shared-drive representative rules. The modeling target is aggregated independently and pre-race inputs do not use these outcome rules.



Exact notebook export: 1_4_dedup_rules.png | Run 20261018T210548Z

Figure A12 | 1 4 participation evidence

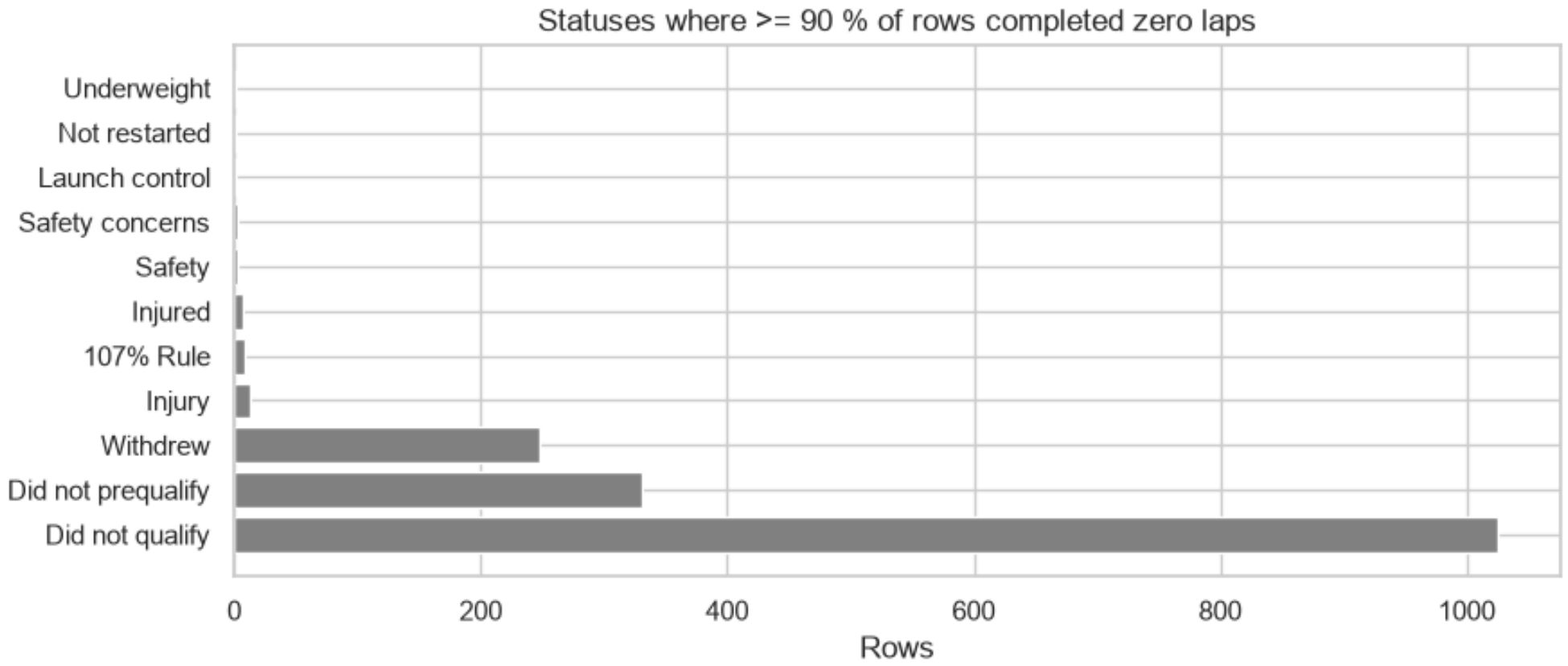
Participation evidence for recorded entries. Zero-lap ambiguity is preserved in sensitivity analyses instead of treating all such records as known nonstarts.



Exact notebook export: 1_4_participation_evidence.png | Run 20261018T210548Z

Figure A13 | 1 4 zero lap statuses

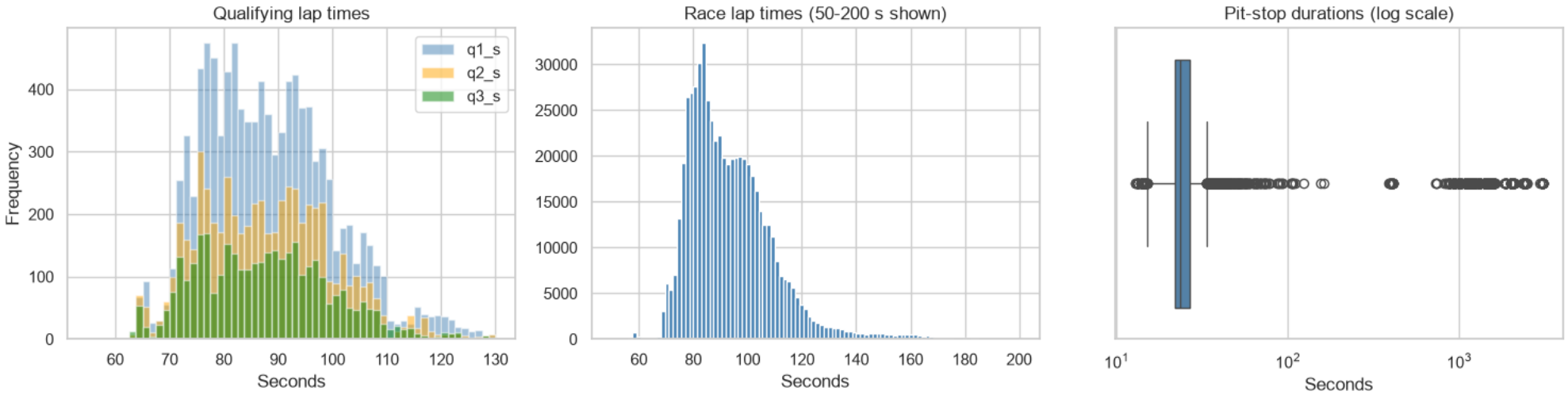
Total result-row counts for status labels whose zero-lap share is at least 90%. The bars count all rows carrying each selected status, including any positive-lap rows.



Exact notebook export: 1_4_zero_lap_statuses.png | Run 20261018T210548Z

Figure A14 | 1 5 durations

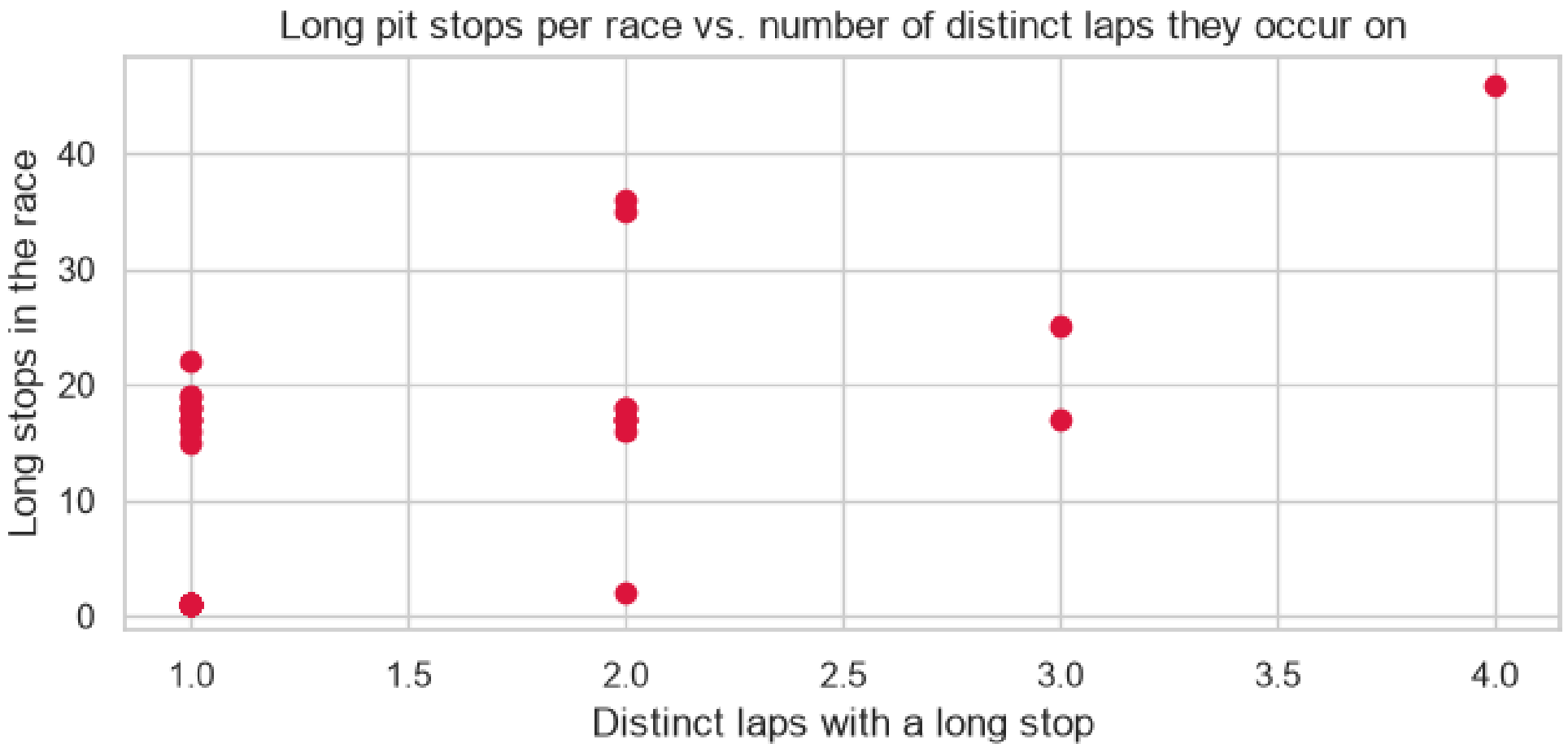
Semantic conversion of duration formats to numeric time. Raw strings are retained for audit; missing or inapplicable durations are not zero time.



Exact notebook export: 1_5_durations.png | Run 20261018T210548Z

Figure A15 | 1 5 long pit stops

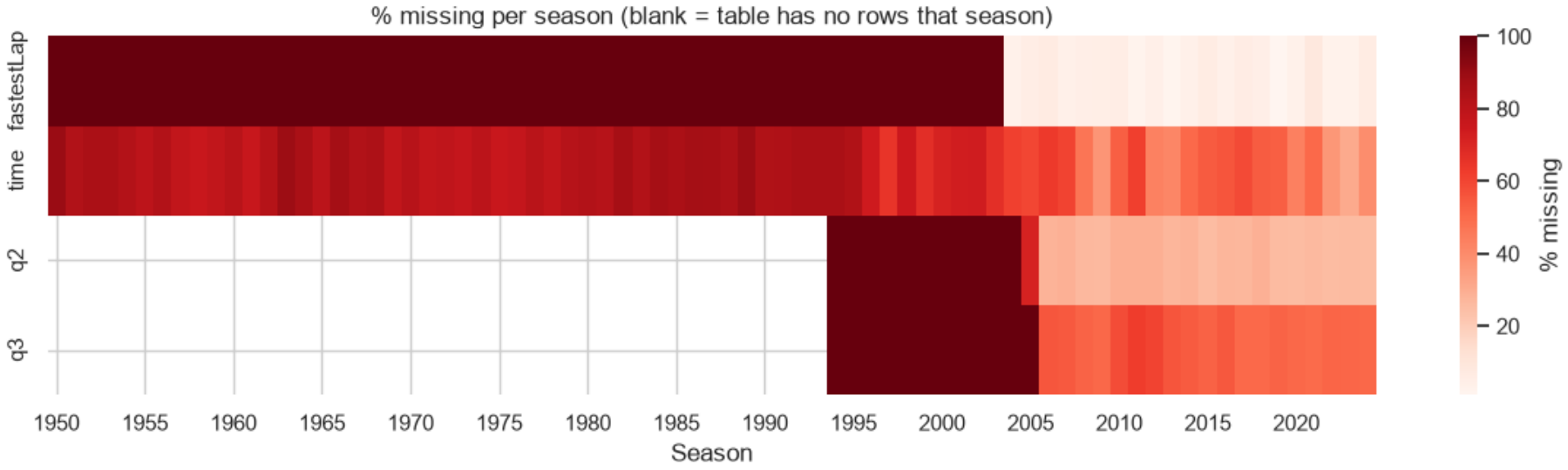
Long pit-stop durations. Extreme recorded times are inspected and flagged rather than removed without task-specific evidence.



Exact notebook export: 1_5_long_pit_stops.png | Run 20261018T210548Z

Figure A16 | 1 5 missing by year

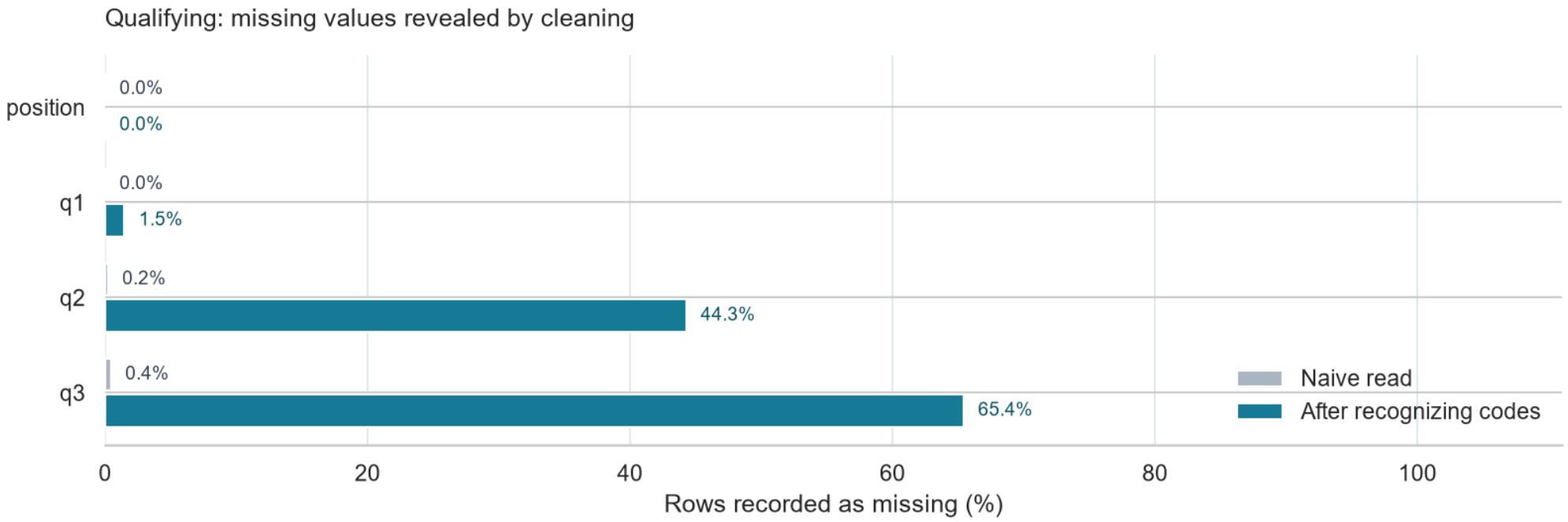
Missingness by year after sentinel conversion. Historical coverage and structural missingness differ from random measurement loss.



Exact notebook export: 1_5_missing_by_year.png | Run 20261018T210548Z

Figure A17 | 1 5 missing qualifying comparison

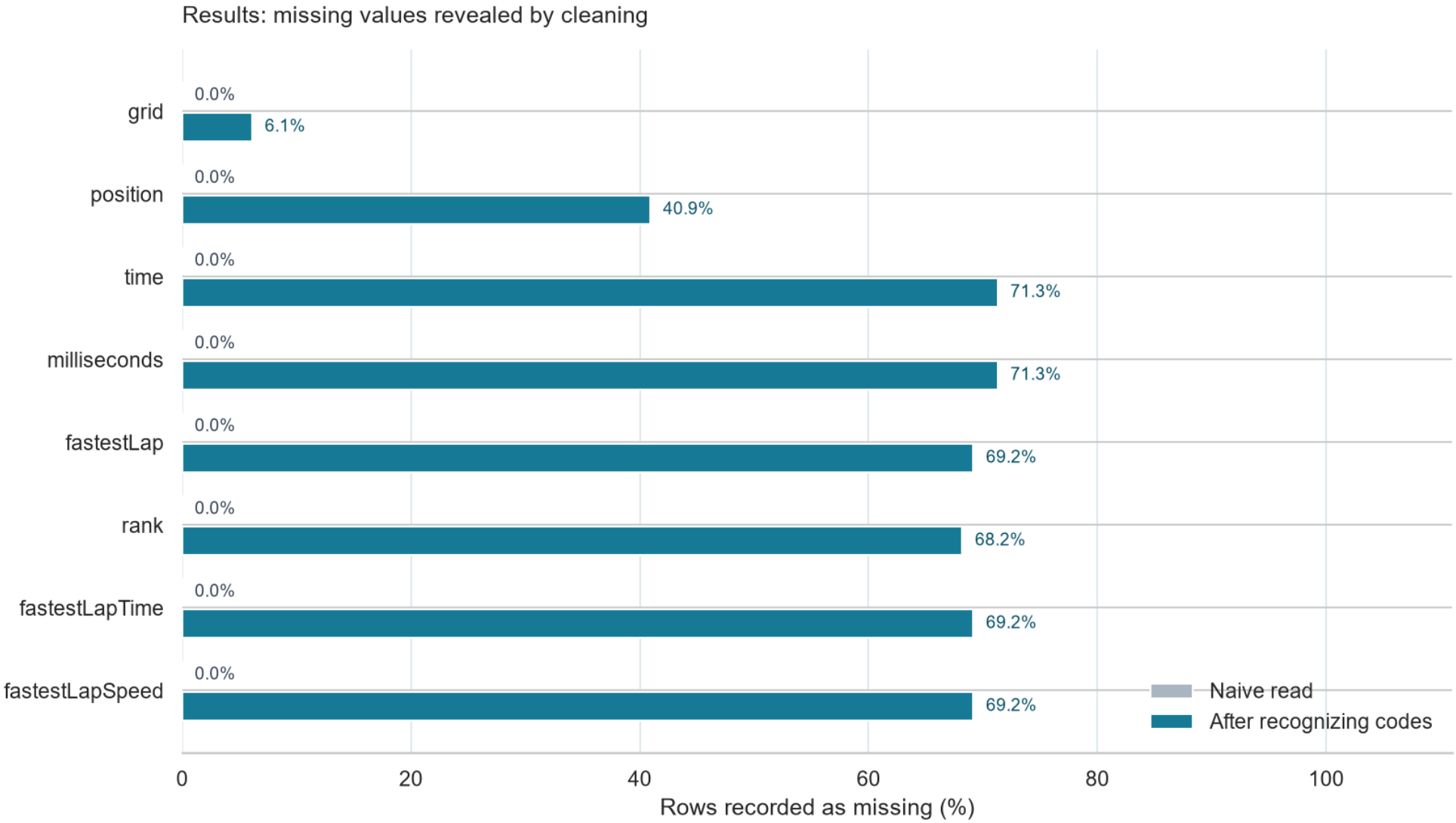
Before/after qualifying missingness evidence. Session progression and historical absence explain why complete-case qualifying modeling would discard useful history.



Exact notebook export: 1_5_missing_qualifying_comparison.png | Run 20261018T210548Z

Figure A18 | 1 5 missing results comparison

Before/after results missingness evidence. Literal sentinel conversion changes apparent null counts; analytical meanings remain field-specific.



Exact notebook export: 1_5_missing_results_comparison.png | Run 20261018T210548Z

Figure A19 | 1 5 missingno matrix

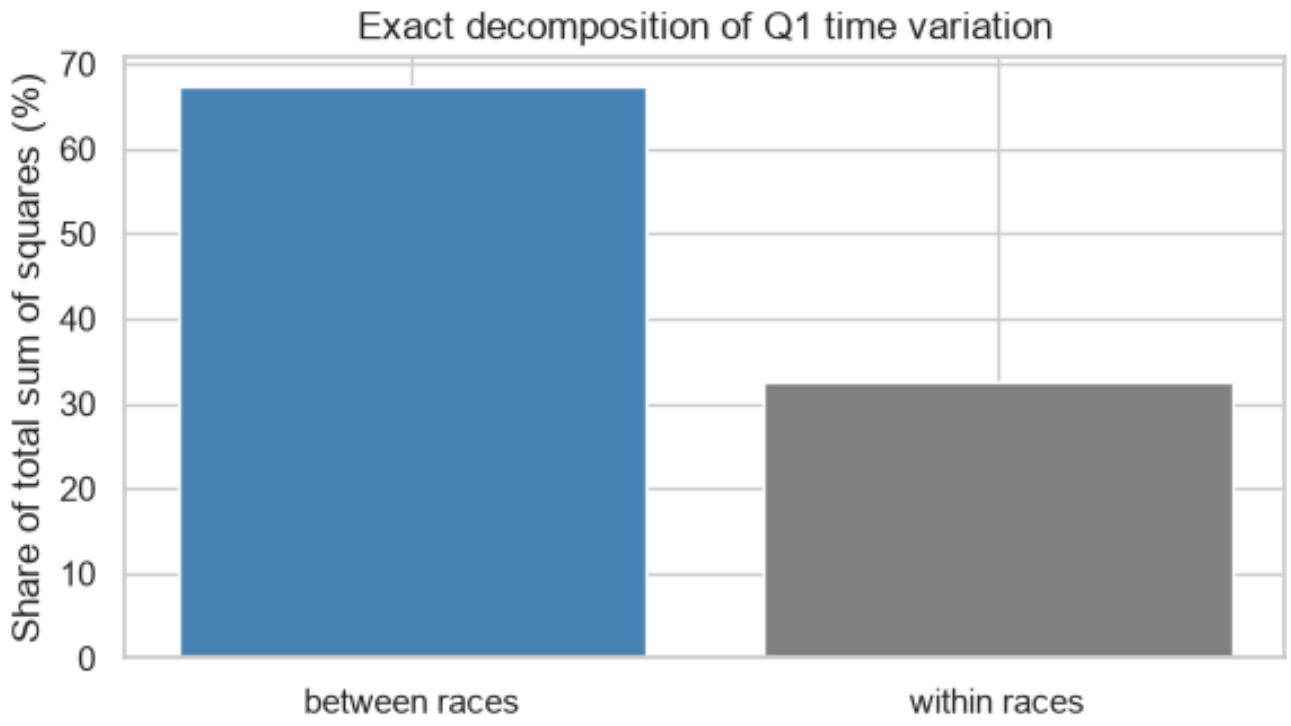
Missingness matrix for selected cleaned result fields, including post-race outcome fields. This diagnostic does not define the prediction schema; no full-period imputation statistics are fitted from it.



Exact notebook export: 1_5_missingno_matrix.png | Run 20261018T210548Z

Figure A20 | 1 5 q1 variance decomposition

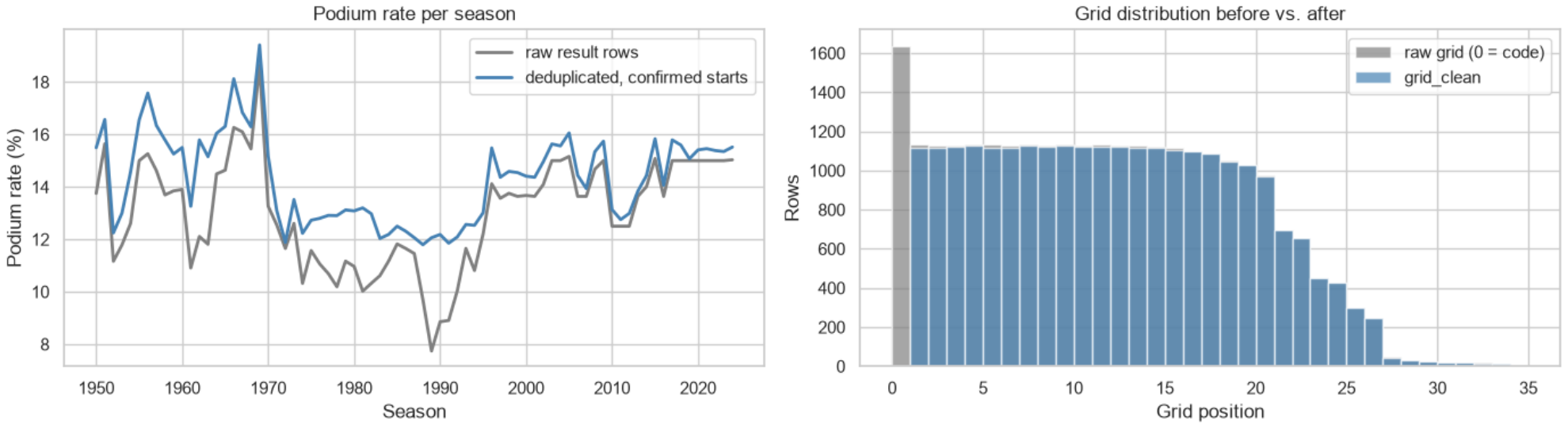
Qualifying-time variation decomposition. Variation across race context limits interpreting raw time as directly comparable performance.



Exact notebook export: 1_5_q1_variance_decomposition.png | Run 20261018T210548Z

Figure A21 | 1 5 target and grid

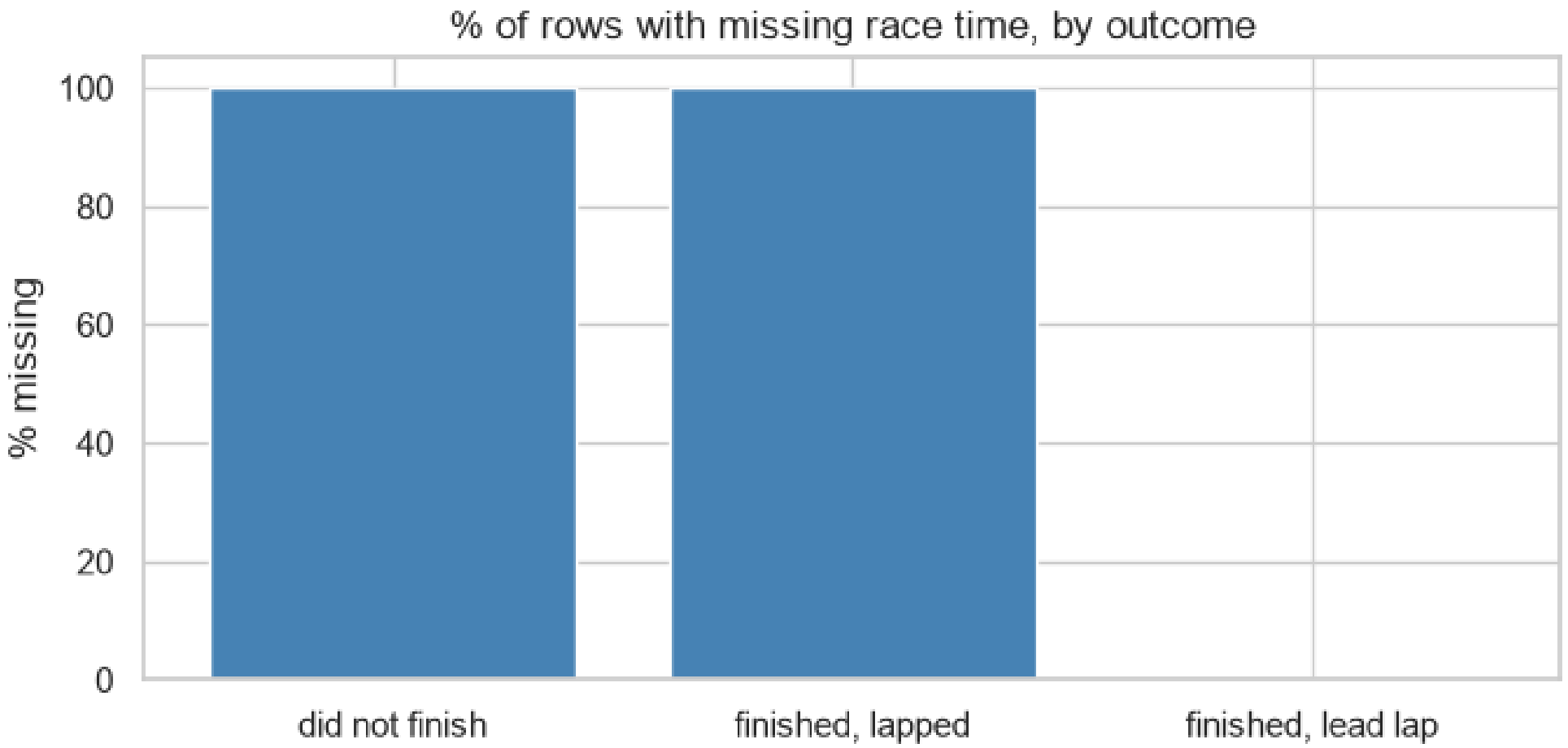
Target and grid distributions after cleaning. Compare with the raw EDA and retain grid-zero information explicitly.



Exact notebook export: 1_5_target_and_grid.png | Run 20261018T210548Z

Figure A22 | 1 5 time missing by outcome

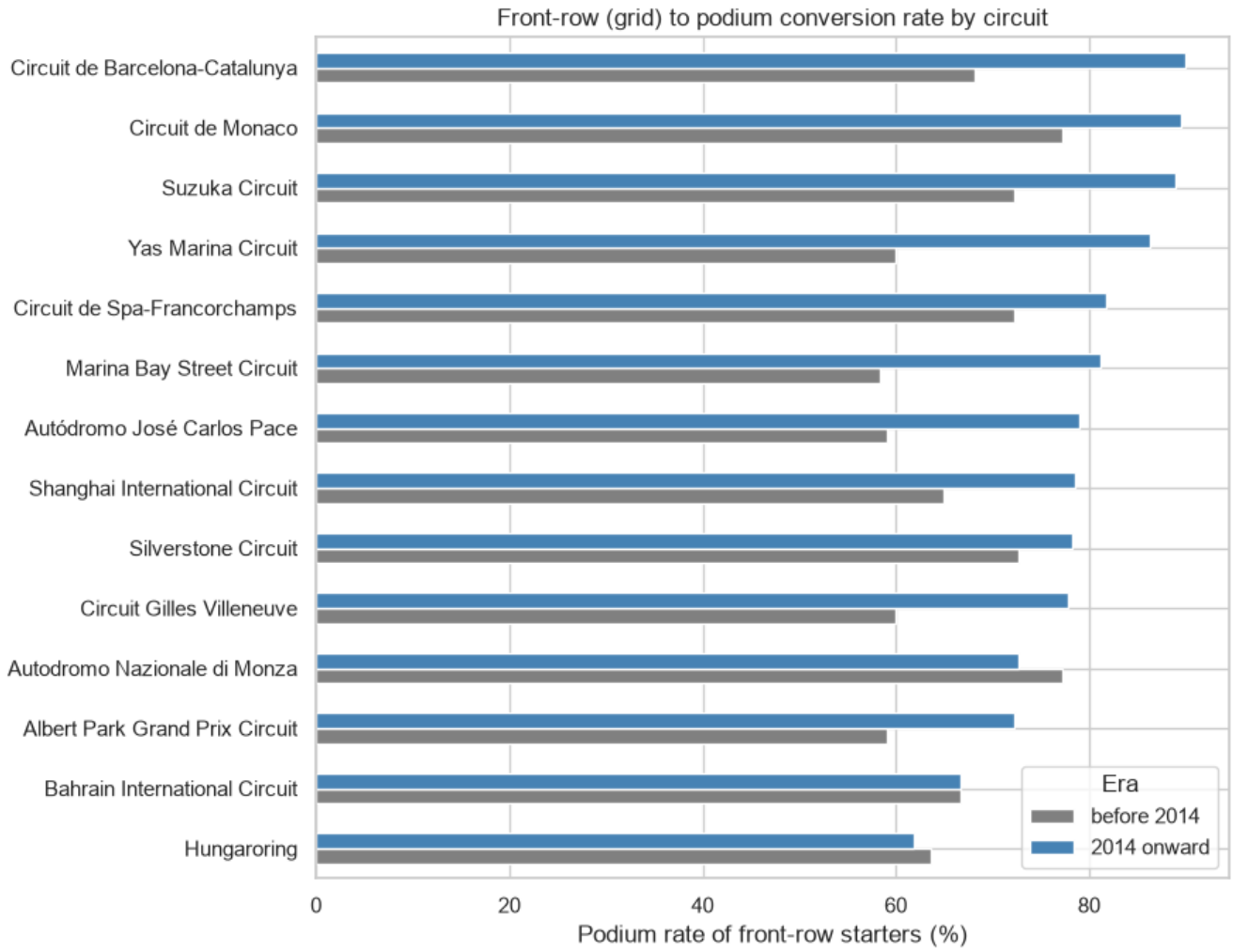
Outcome-dependent missing duration evidence. This supports excluding current-race durations from prediction, not filling them into a pre-race feature.



Exact notebook export: 1_5_time_missing_by_outcome.png | Run 20261018T210548Z

Figure A23 | 2 1 conversion by circuit

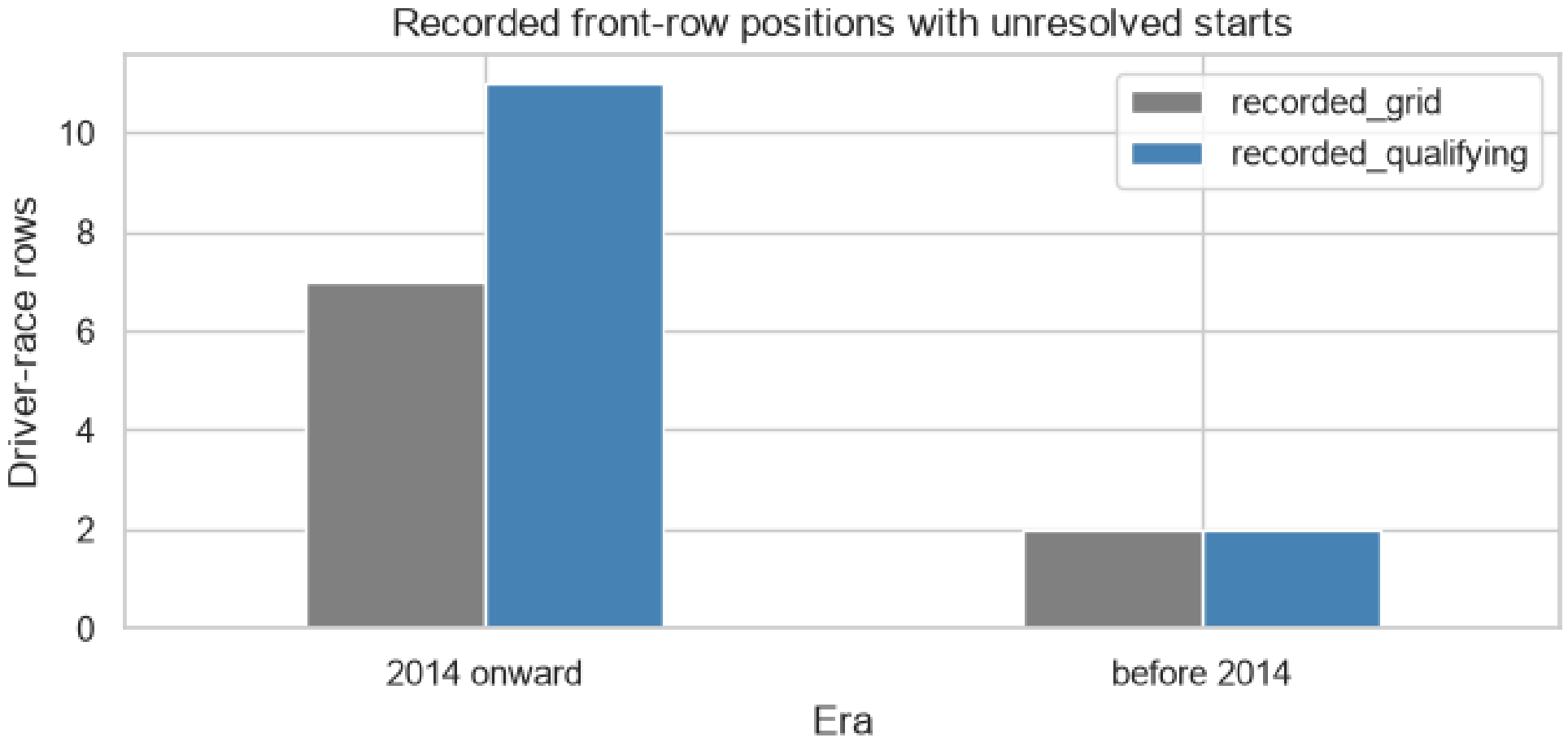
Q1 grid-front-row podium conversion by circuit and era under the main coverage rule. Denominators and participation assumptions qualify the ranking.



Exact notebook export: 2_1_conversion_by_circuit.png | Run 20261018T210548Z

Figure A24 | 2 1 front row uncertainty

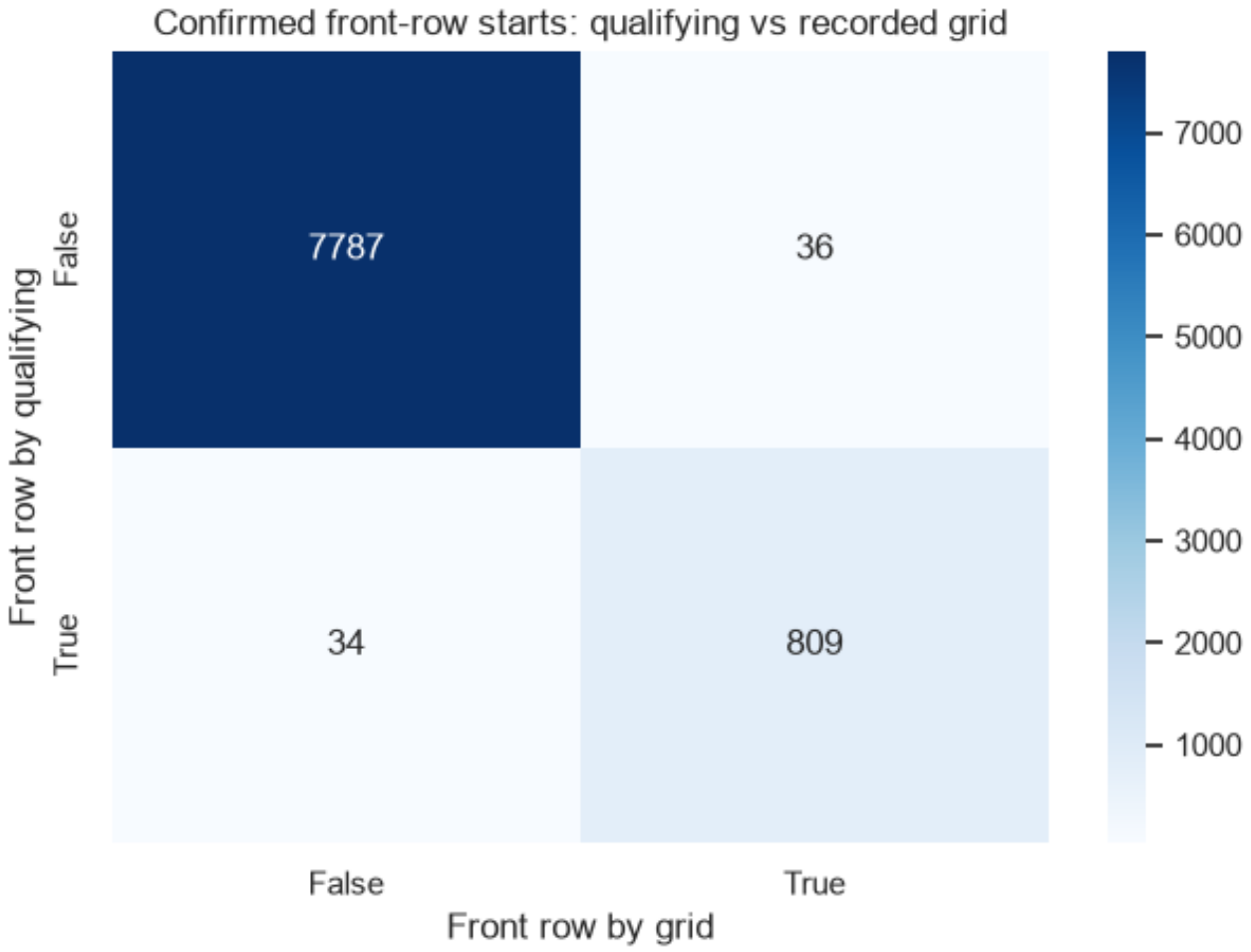
Q1 counts of recorded front-row positions with unresolved zero-lap participation, separated by qualifying/grid definition and era. These entries motivate the rate sensitivity bounds in the exported tables.



Exact notebook export: 2_1_front_row_uncertainty.png | Run 20261018T210548Z

Figure A25 | 2 1 quali vs grid

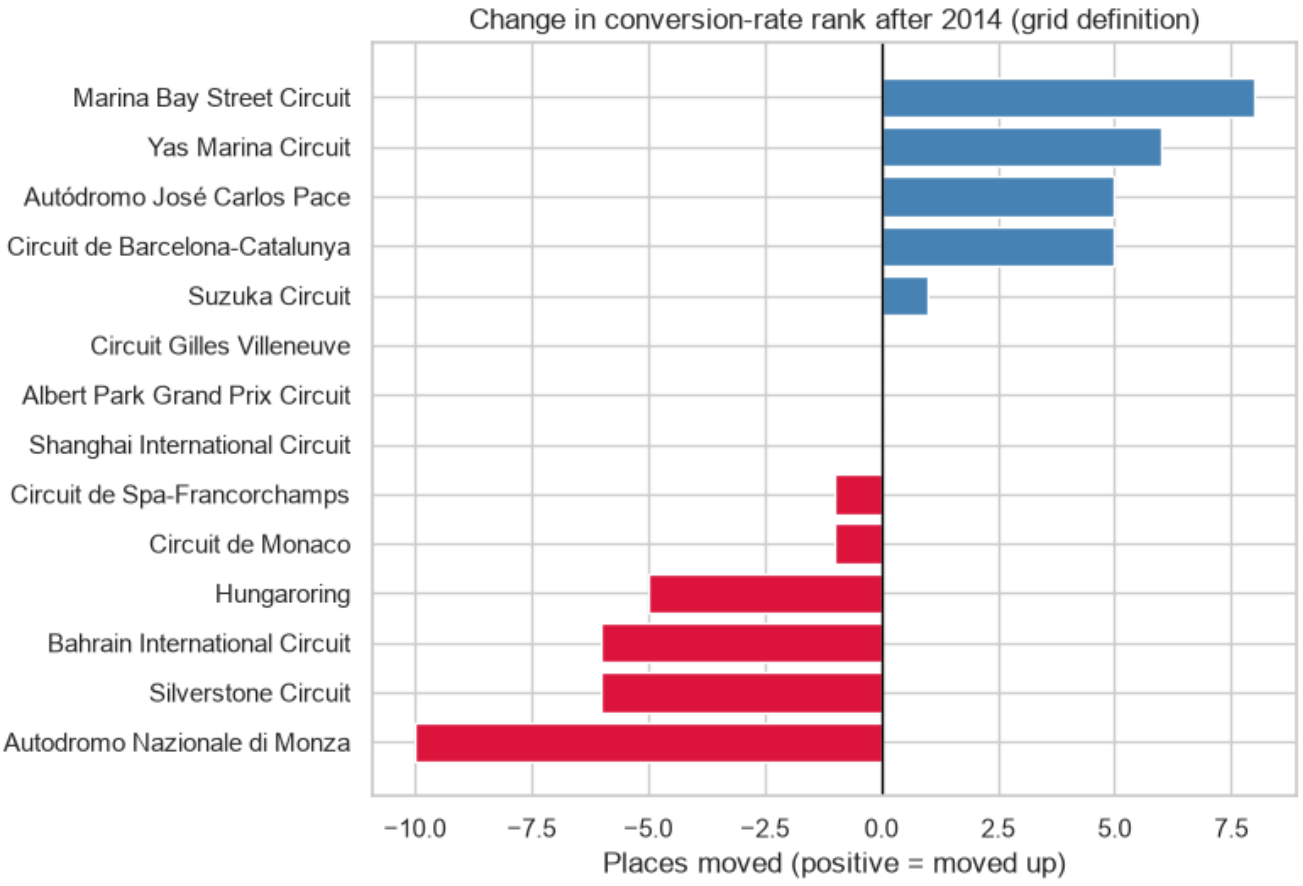
Q1 agreement among 8,666 positive-lap confirmed starters only. Unresolved zero-lap entries are excluded. Penalties and substitutions can change front-row membership between qualifying and recorded grid.



Exact notebook export: 2_1_quali_vs_grid.png | Run 20261018T210548Z

Figure A26 | 2 1 rank change

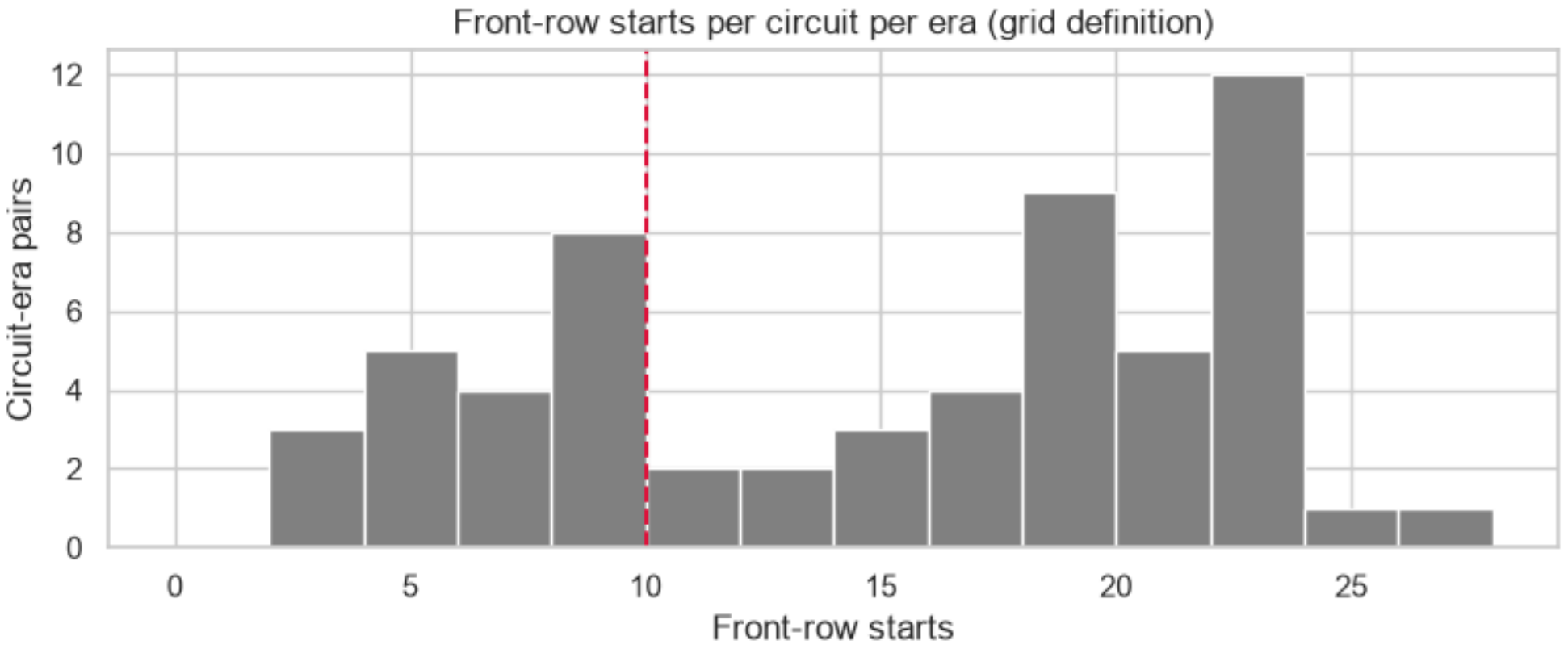
Q1 era ranking changes within the common eligible circuits. Rankings reflect composition and sample sizes as well as circuit differences.



Exact notebook export: 2_1_rank_change.png | Run 20261018T210548Z

Figure A27 | 2 1 sample size

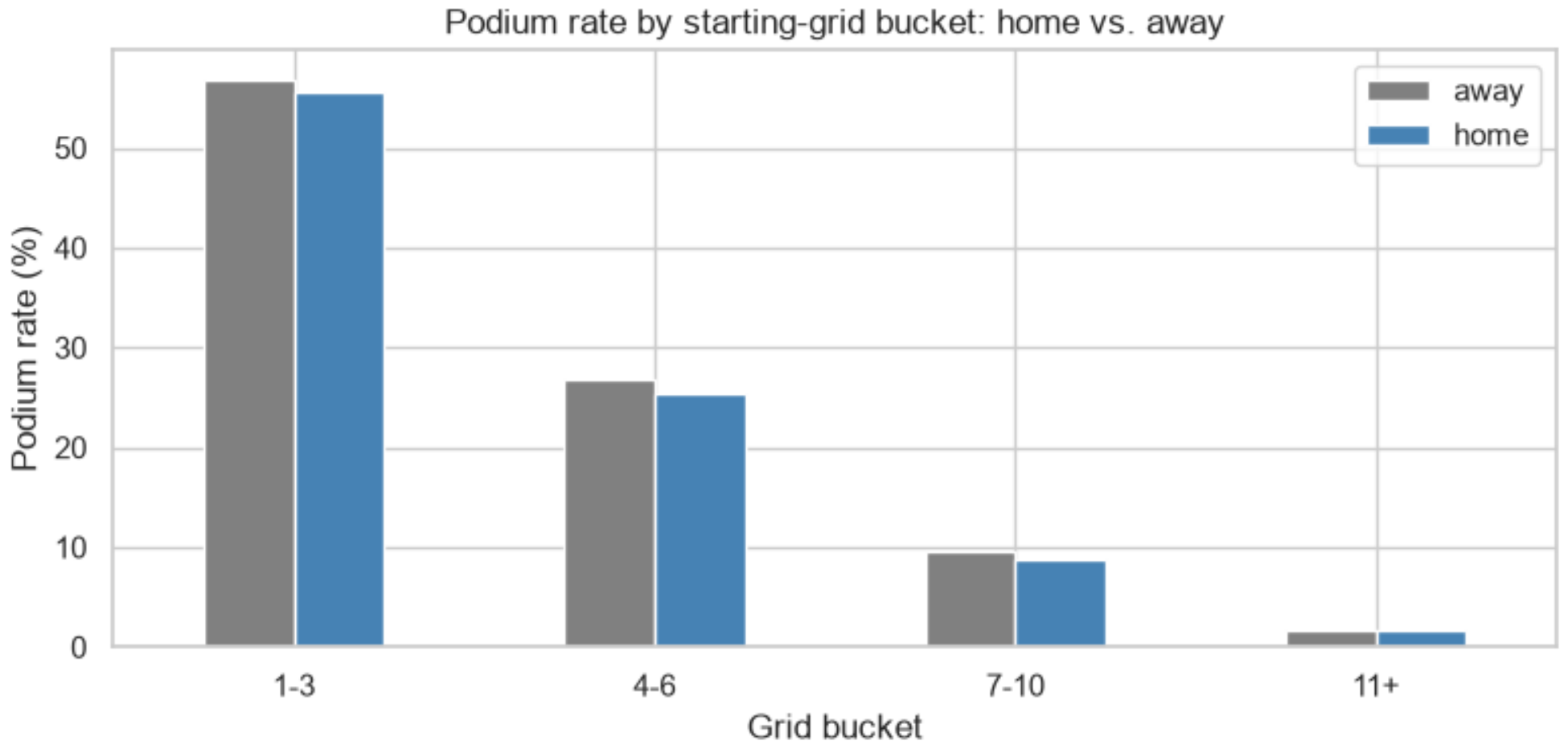
Distribution of confirmed front-row counts across circuit-era pairs in 2003-2024. The dashed line marks 10 starts. This histogram motivates the common-circuit cutoff; it does not display unrestricted conversion rates.



Exact notebook export: 2_1_sample_size.png | Run 20261018T210548Z

Figure A28 | 2 2 home vs away

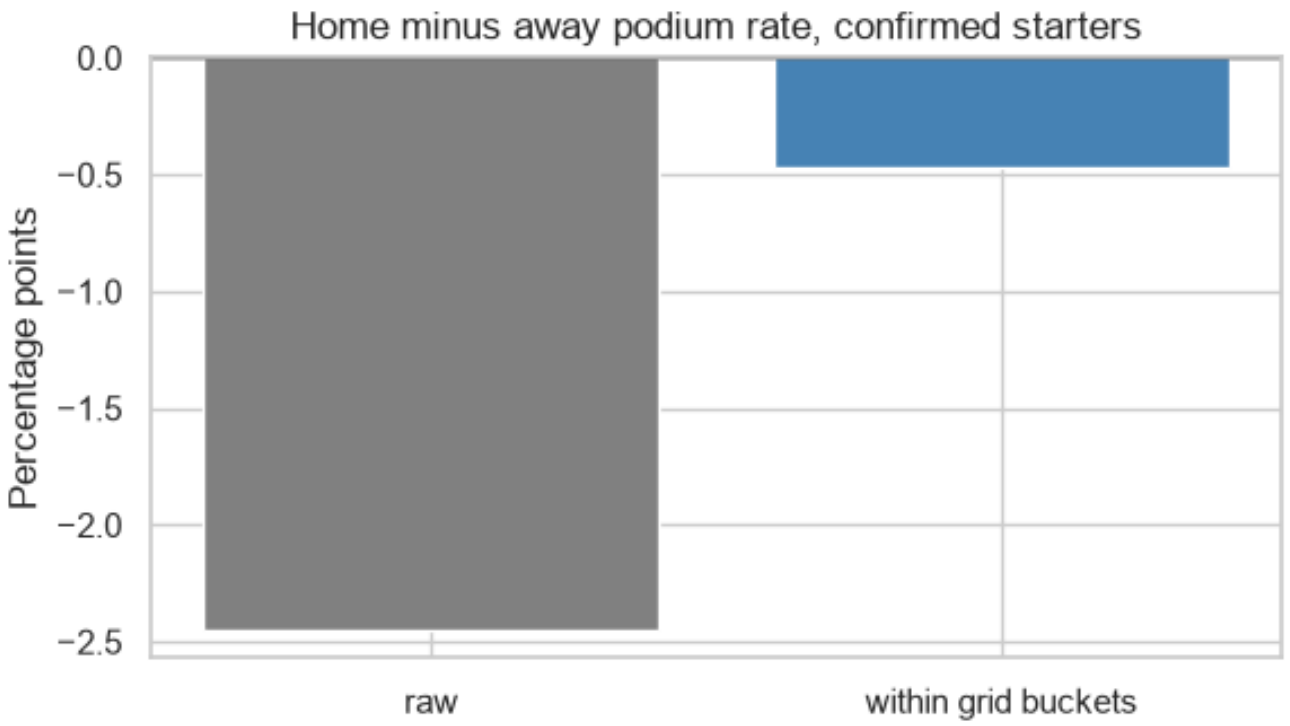
Q2 home versus away podium rates within common grid buckets. This adjusts broad starting-position composition only.



Exact notebook export: 2_2_home_vs_away.png | Run 20261018T210548Z

Figure A29 | 2 2 raw vs controlled

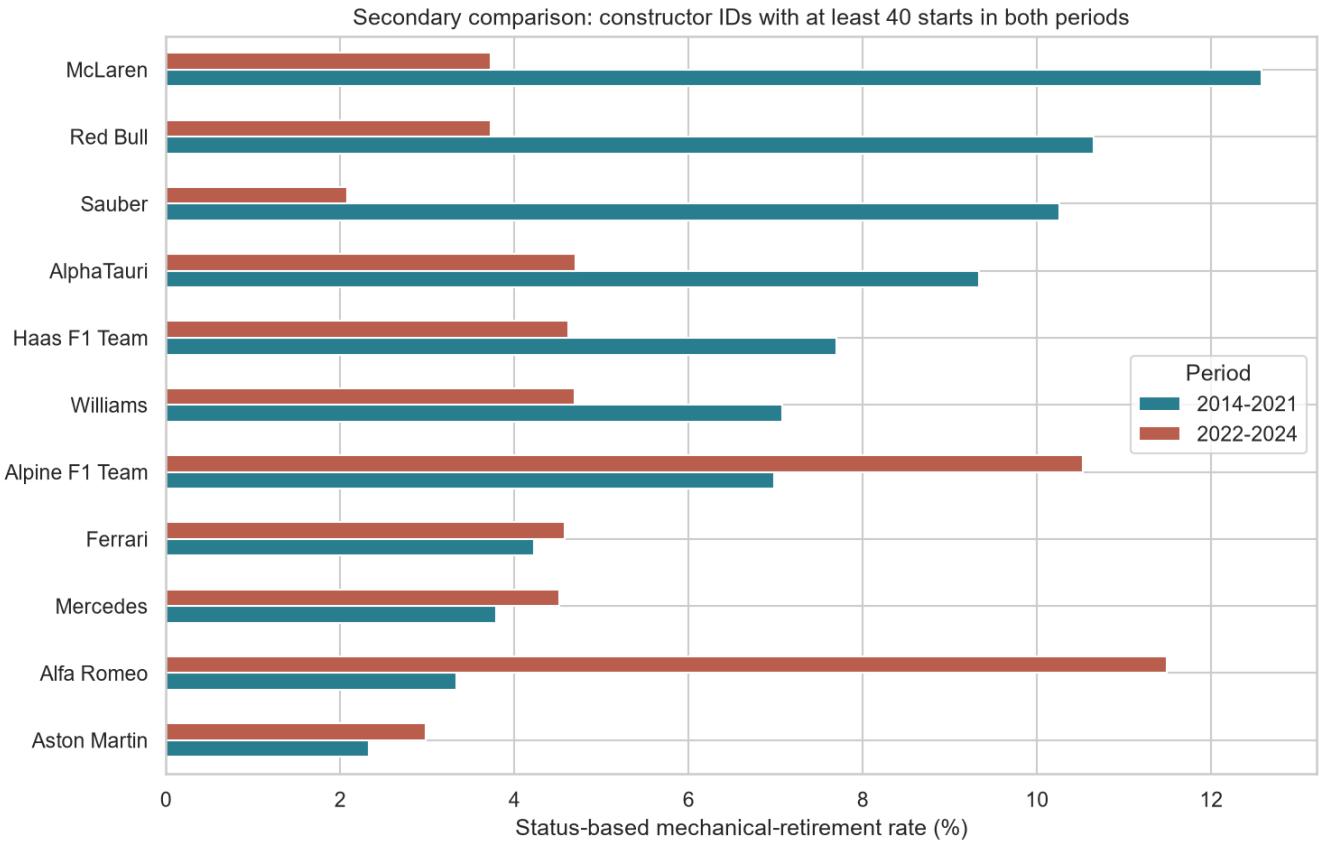
Q2 raw and grid-weighted home-minus-away differences among confirmed starters. The separate written answer reports the unresolved-start sensitivity. These are descriptive associations, not causal home effects.



Exact notebook export: 2_2_raw_vs_controlled.png | Run 20261018T210548Z

Figure A30 | 2 3 matched constructor comparison

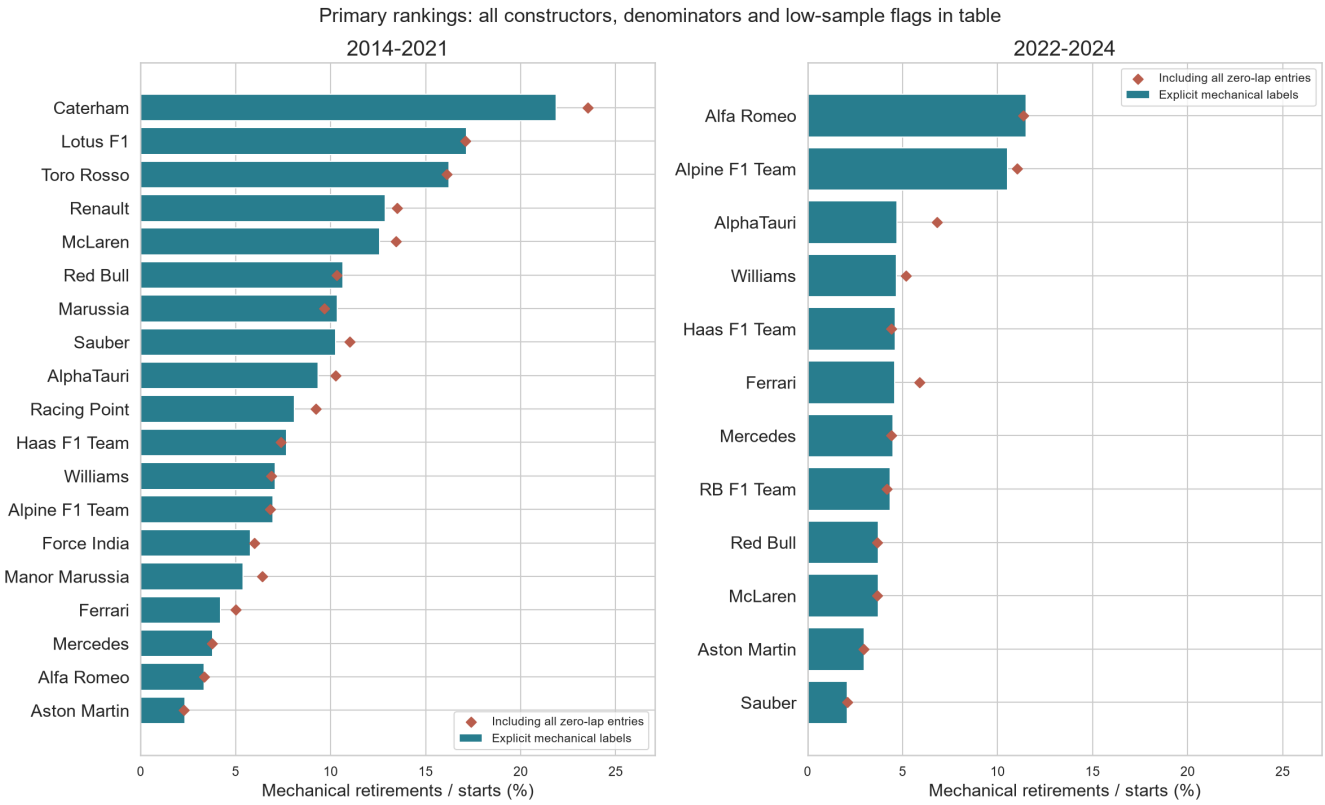
Q3 secondary same-ID comparison with >=40 confirmed starts in both periods. Constructor rebrands are not merged into a lineage.



Exact notebook export: 2_3_matched_constructor_comparison.png | Run 20261018T210548Z

Figure A31 | 2 3 mechanical retirements

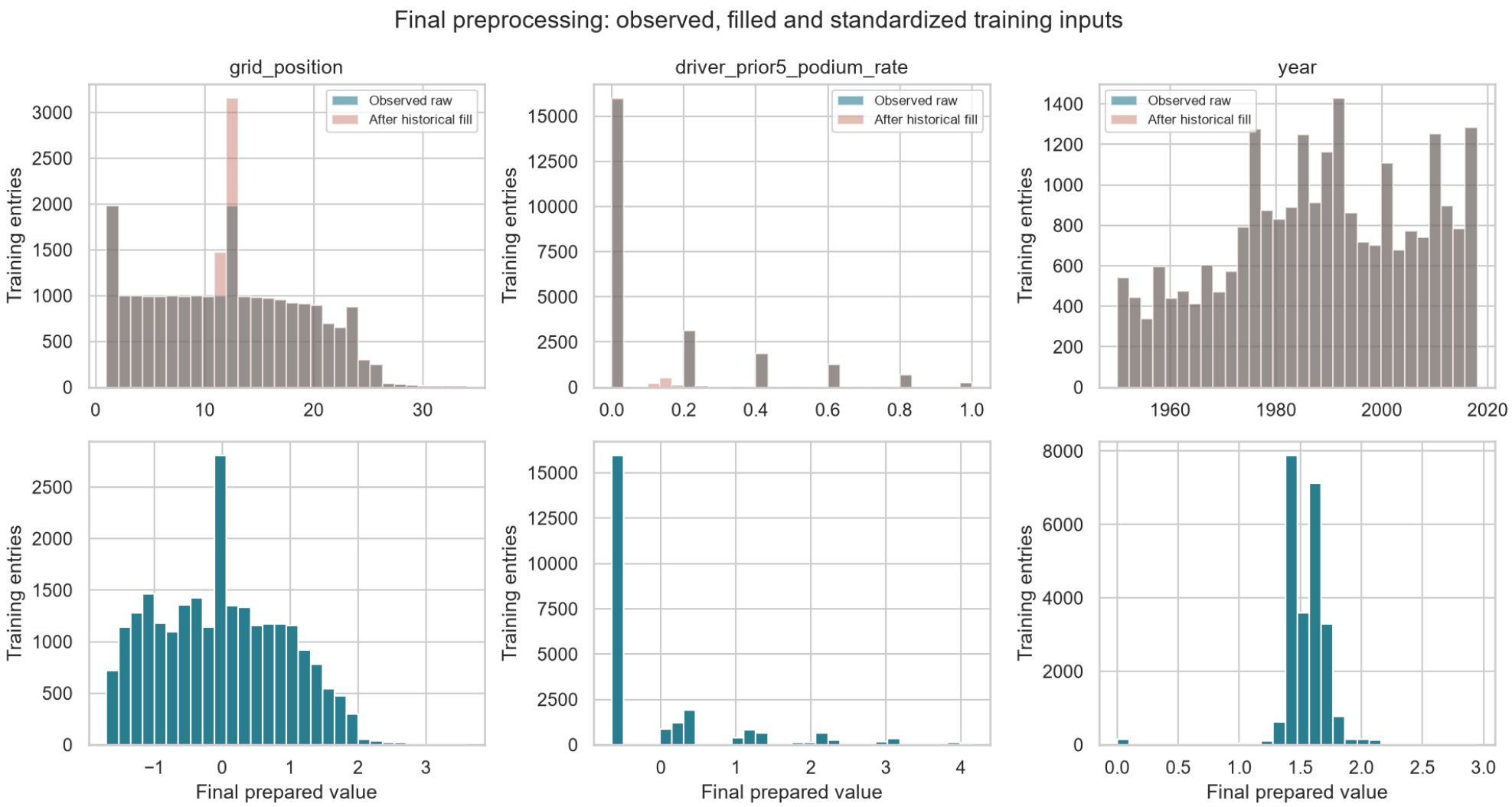
Q3 primary mechanical-retirement ranking includes every constructor. Zero-lap sensitivity is shown alongside the explicit-cause confirmed-start rate; table denominators flag small samples.



Exact notebook export: 2_3_mechanical_retirements.png | Run 20261018T210548Z

Figure A32 | 3 final preprocessing values

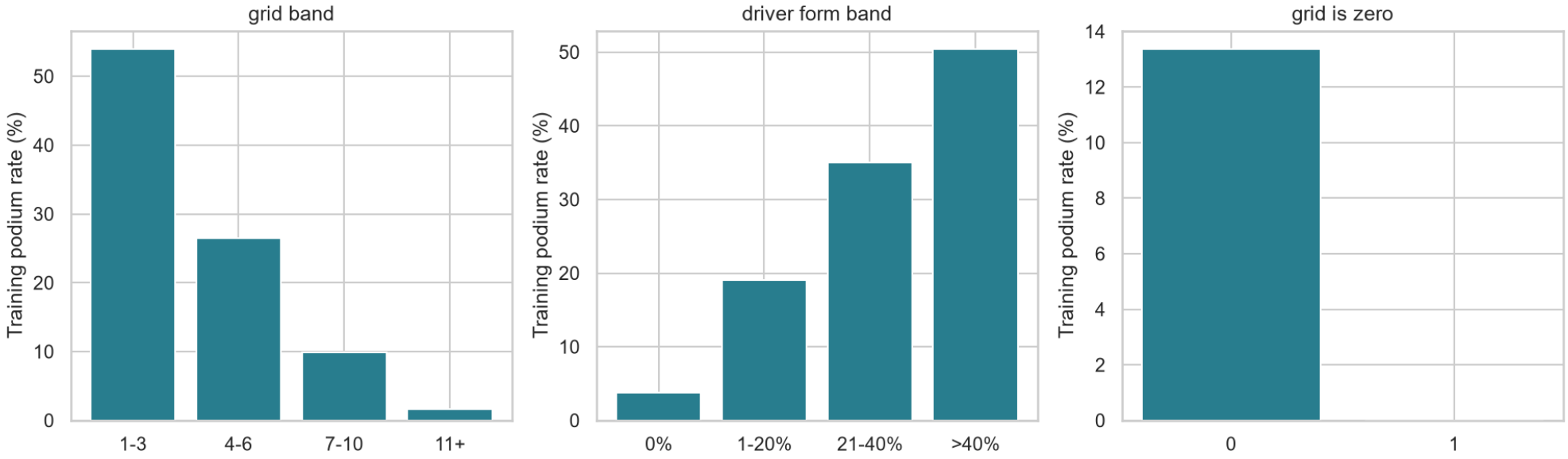
Direct before/after evidence for the selected common pipeline: observed, historical-fill and prepared values. Statistics use strictly earlier race dates.



Exact notebook export: 3_final_preprocessing_values.png | Run 20261018T210548Z

Figure A33 | 3 training feature evidence

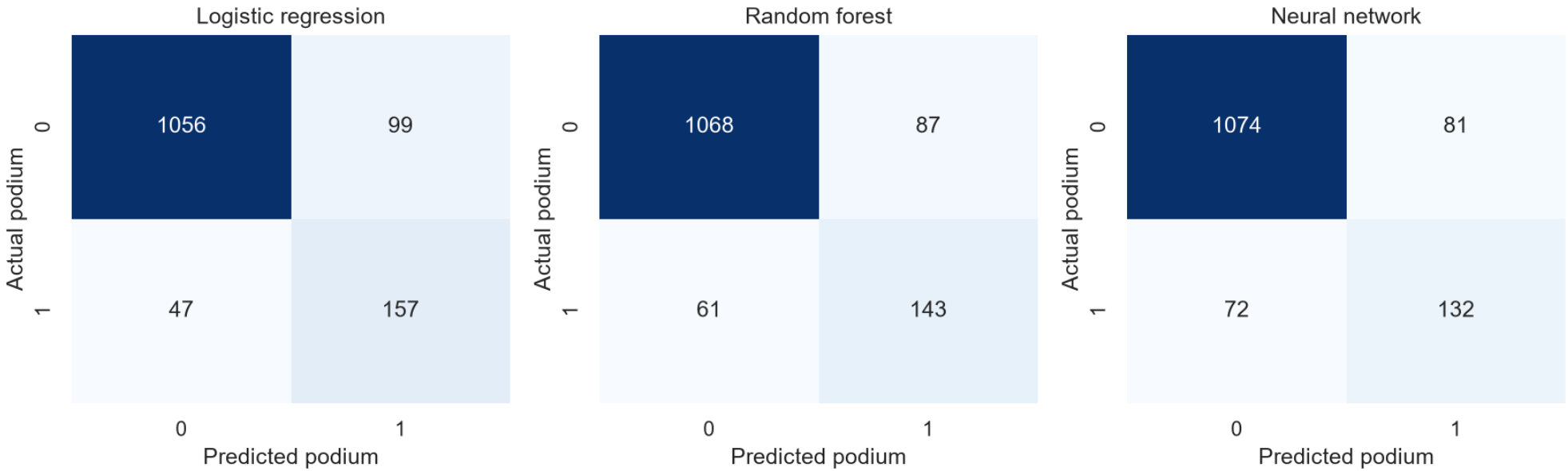
Training-only EDA motivating grid, driver form and grid-zero information. Later ablations test conditional contribution for all model families.



Exact notebook export: 3_training_feature_evidence.png | Run 20261018T210548Z

Figure A34 | 4 confusion all models

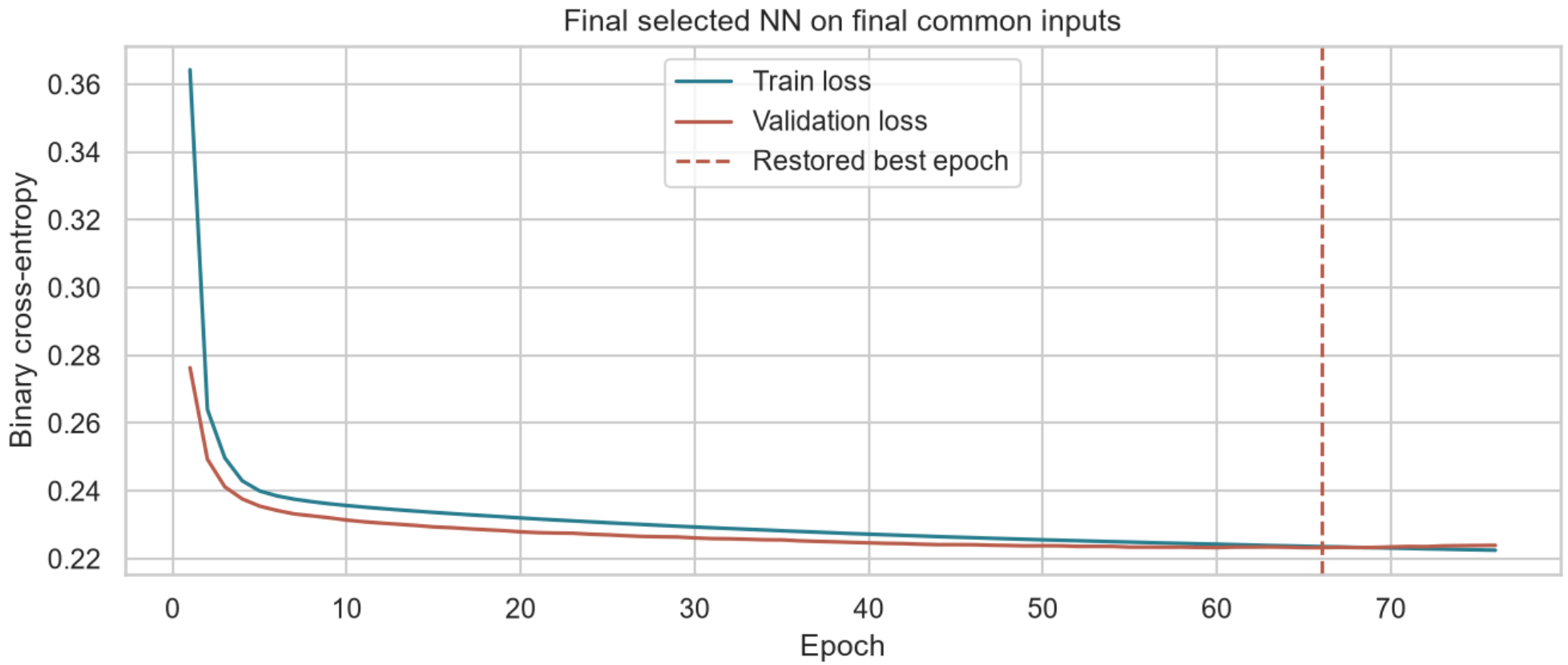
Final confusion matrices for all three families at validation-selected thresholds. Differences represent the selected operating points on the same test population.



Exact notebook export: 4_confusion_all_models.png | Run 20261018T210548Z

Figure A35 | 4 final nn learning curve

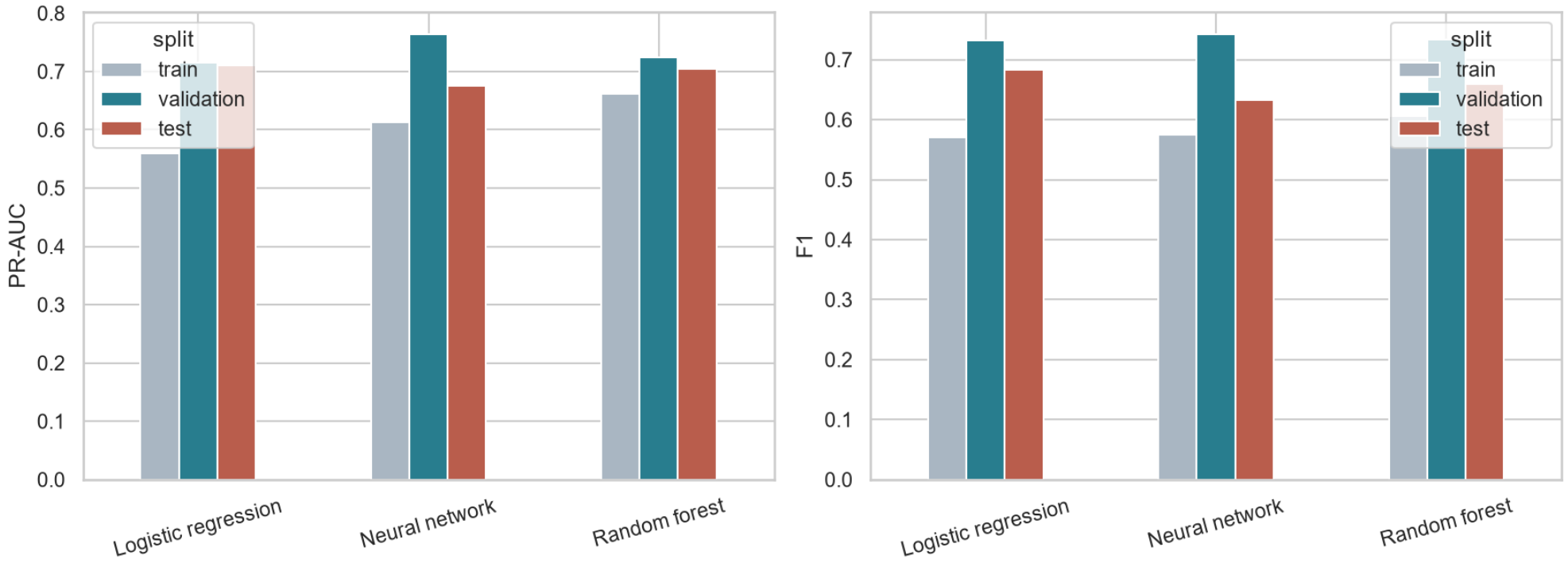
Final shallow-network learning curve on selected inputs and rows. Best validation-loss weights are restored; the exported summary gives run and best epochs.



Exact notebook export: 4_final_nn_learning_curve.png | Run 20261018T210548Z

Figure A36 | 4 model comparison

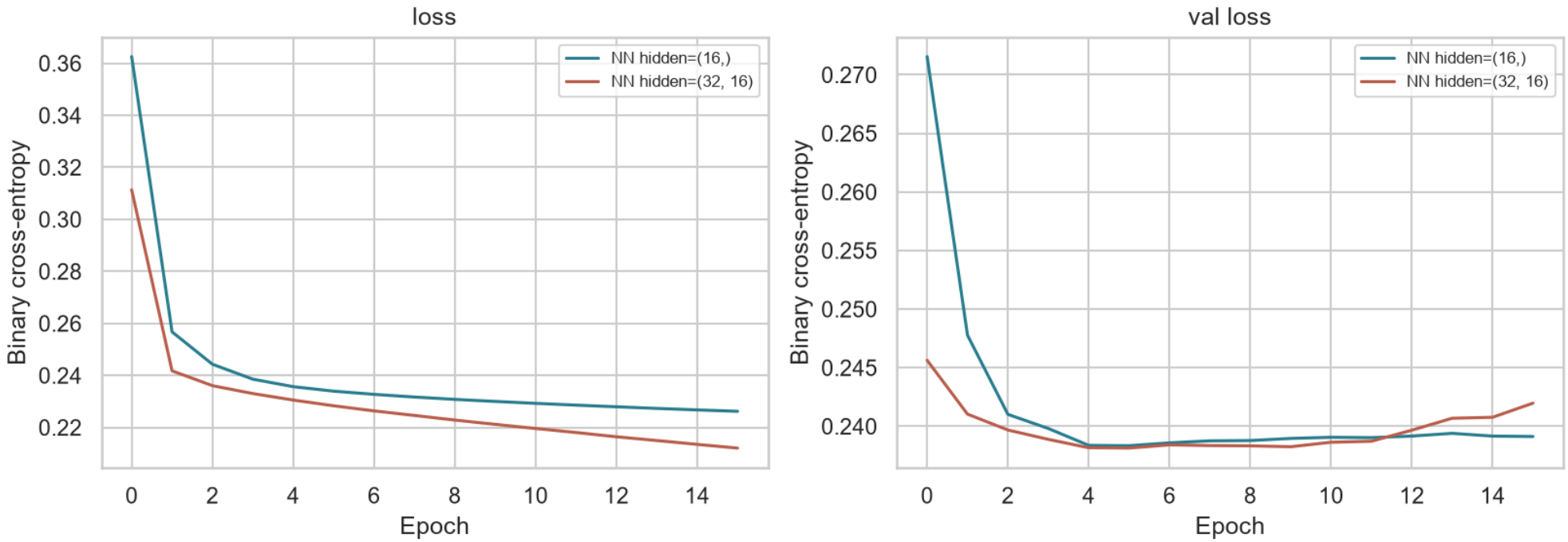
Final train/validation/test average precision and F1 for the three families. Final input rows and columns are identical; test was viewed in prior revisions.



Exact notebook export: 4_model_comparison.png | Run 20261018T210548Z

Figure A37 | 4 nn learning curves

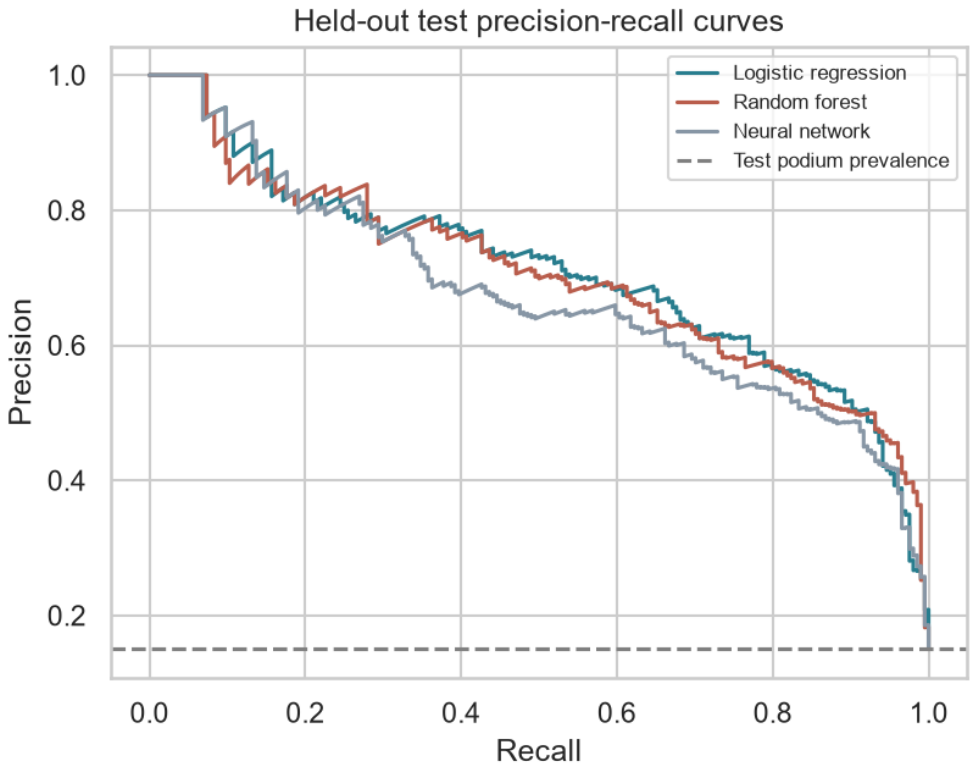
Reference neural-network training and validation learning curves. These precede common pipeline selection; final-model curves are reported separately.

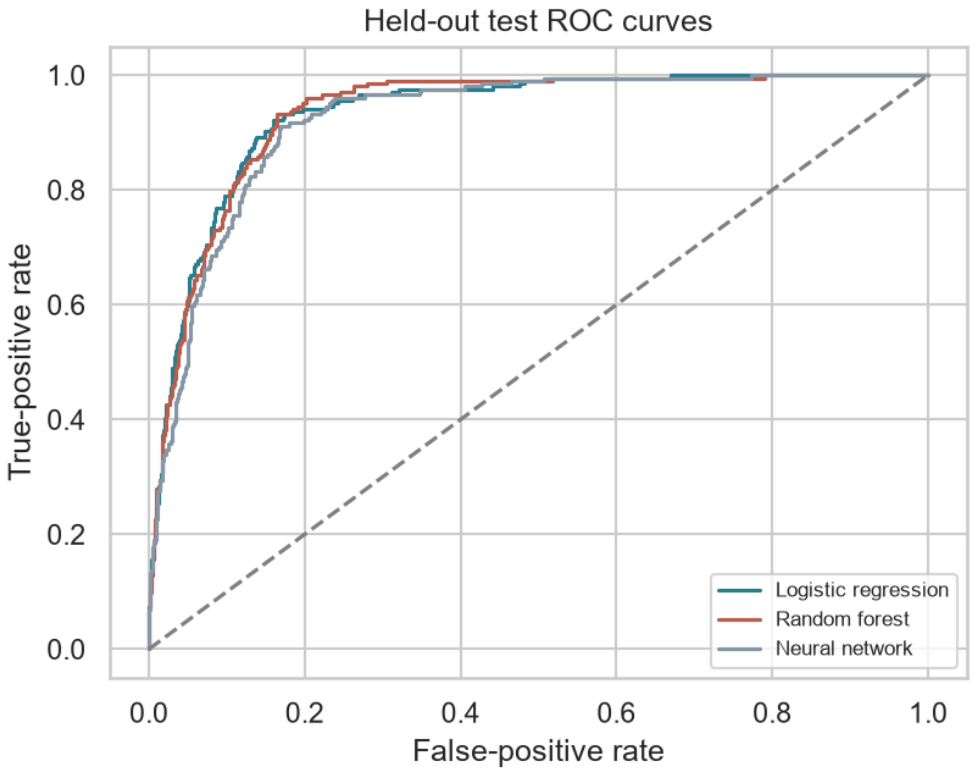


Exact notebook export: 4_nn_learning_curves.png | Run 20261018T210548Z

Figure A38 | 4 test curves

Final frozen-model ROC and precision-recall curves on the revised later-era test evaluation. No threshold or hyperparameter is selected from these curves.

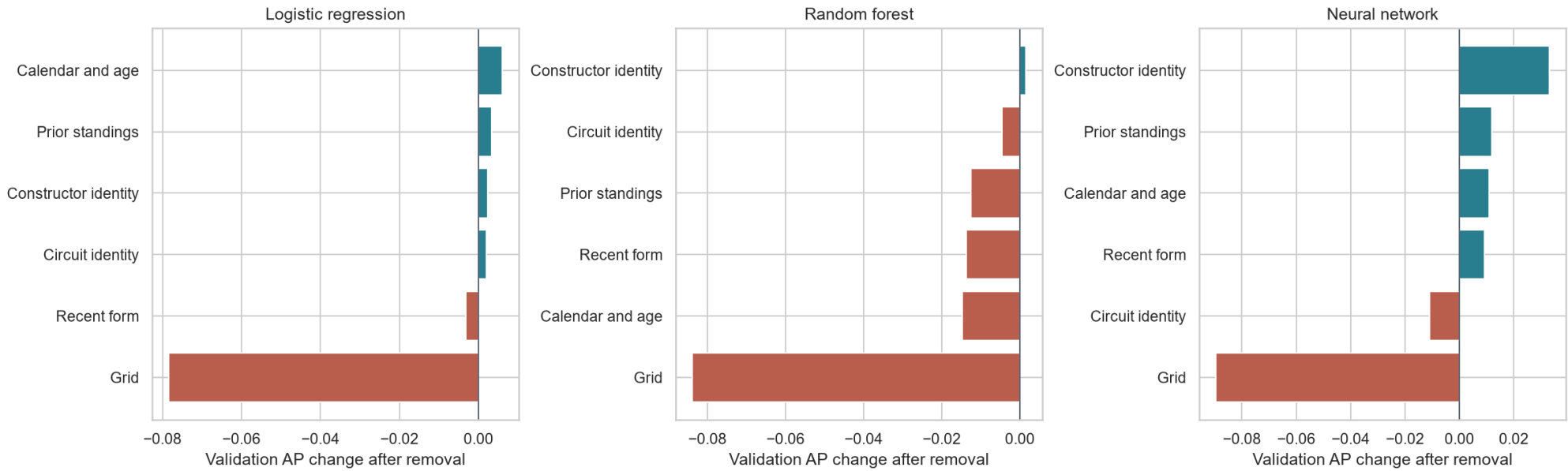




Exact notebook export: 4_test_curves.png | Run 20261018T210548Z

Figure A39 | 5 ablation all models

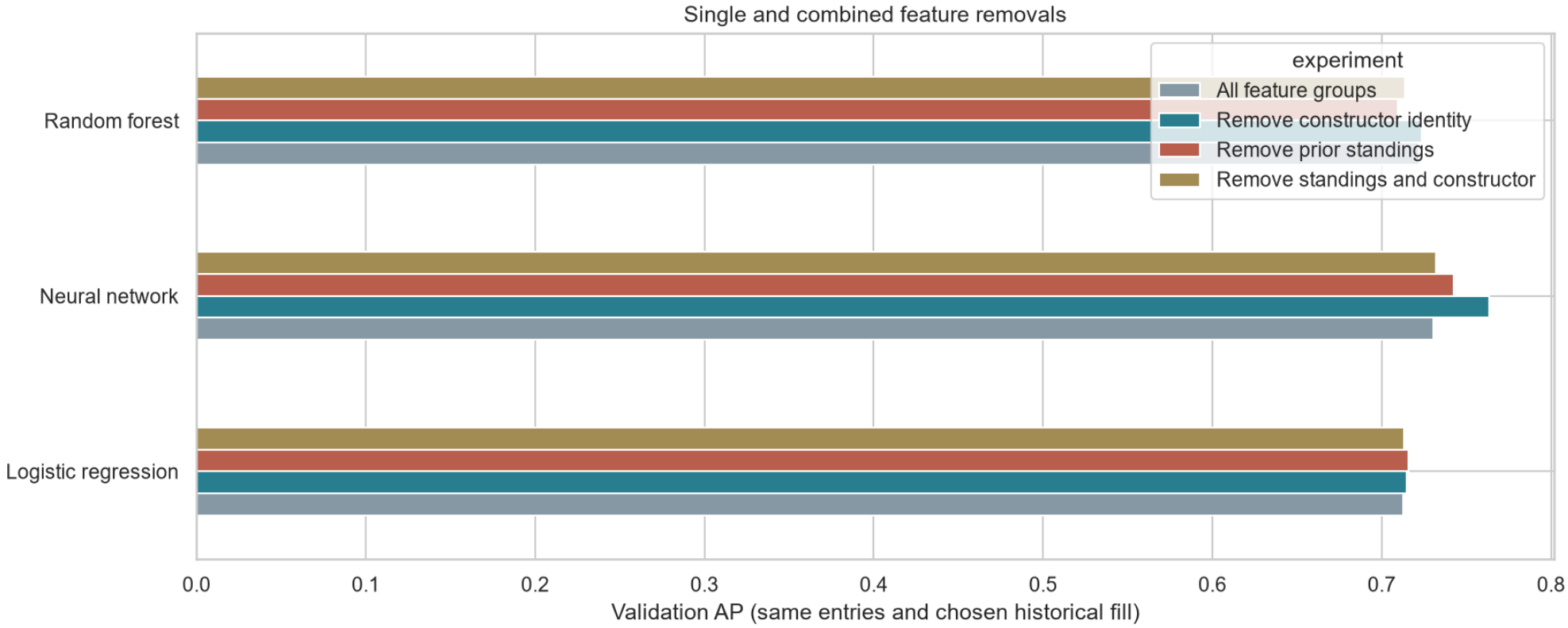
Part 5 retrained feature-group ablations for all families. AP changes are relative to each family's full-feature reference, not to the final reduced pipeline.



Exact notebook export: 5_ablation_all_models.png | Run 20261018T210548Z

Figure A40 | 5 combined feature removals

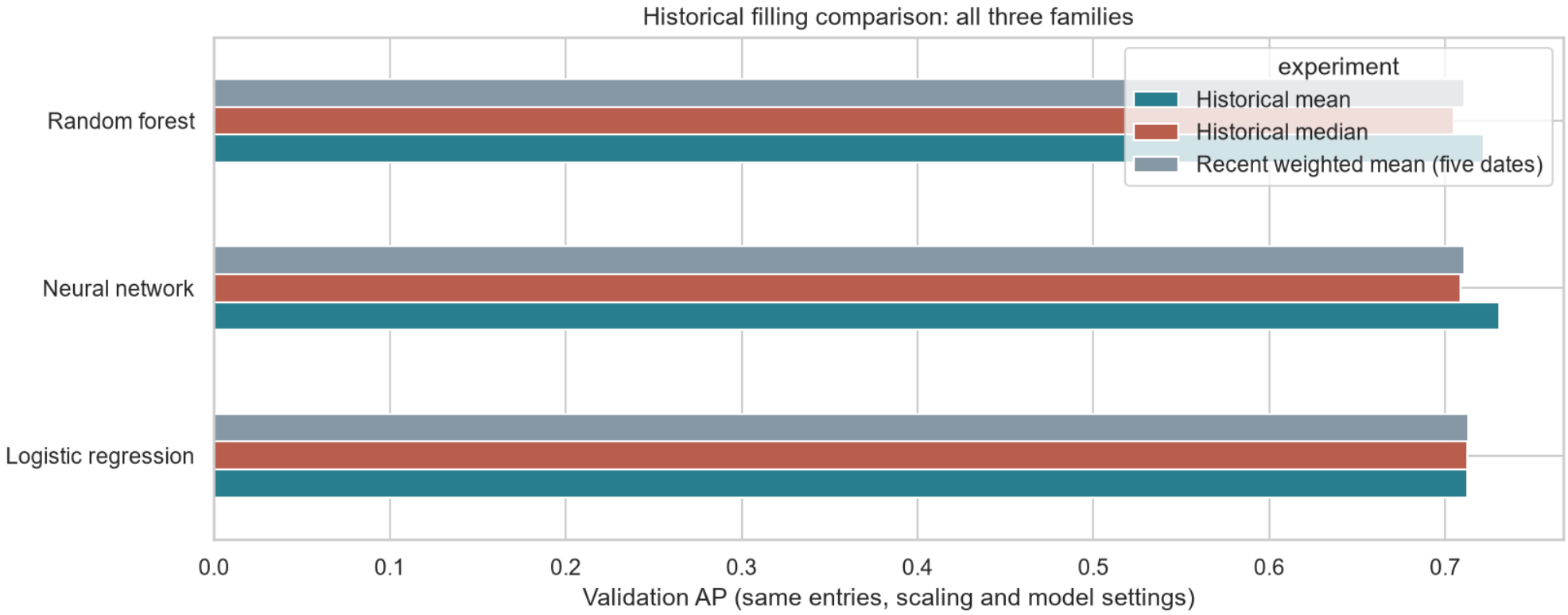
Joint feature-removal candidates evaluated separately. Improvements from individual removals cannot be added without this combined fit.



Exact notebook export: 5_combined_feature_removals.png | Run 20261018T210548Z

Figure A41 | 5 historical fills

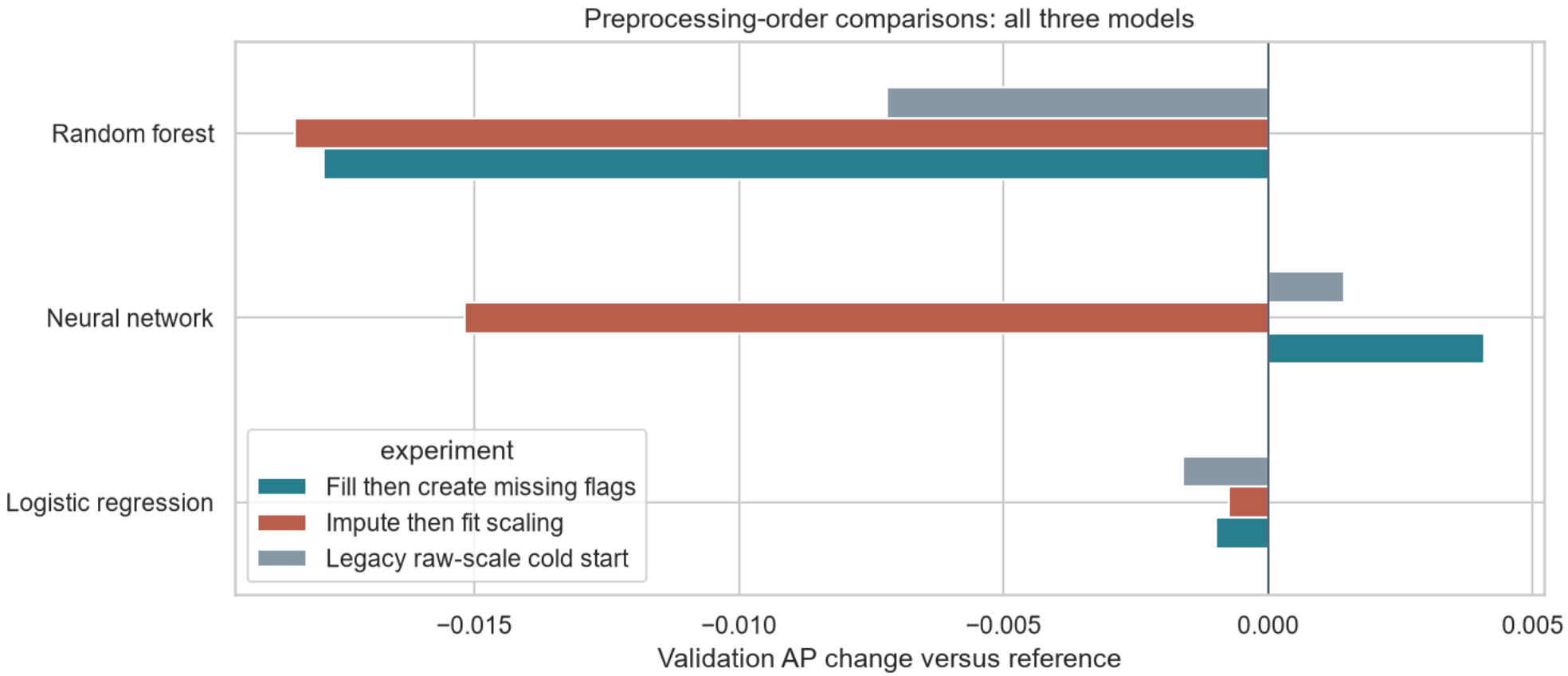
Historical mean, median and recent weighted filling for all families. Every candidate uses only strictly earlier race dates.



Exact notebook export: 5_historical_fills.png | Run 20261018T210548Z

Figure A42 | 5 preprocessing all models

Part 5 order comparisons and legacy cold-start diagnostic for all families. No values are scrambled in these training experiments.



Exact notebook export: 5_preprocessing_all_models.png | Run 20261018T210548Z

Figure A43 | 5 training row policies

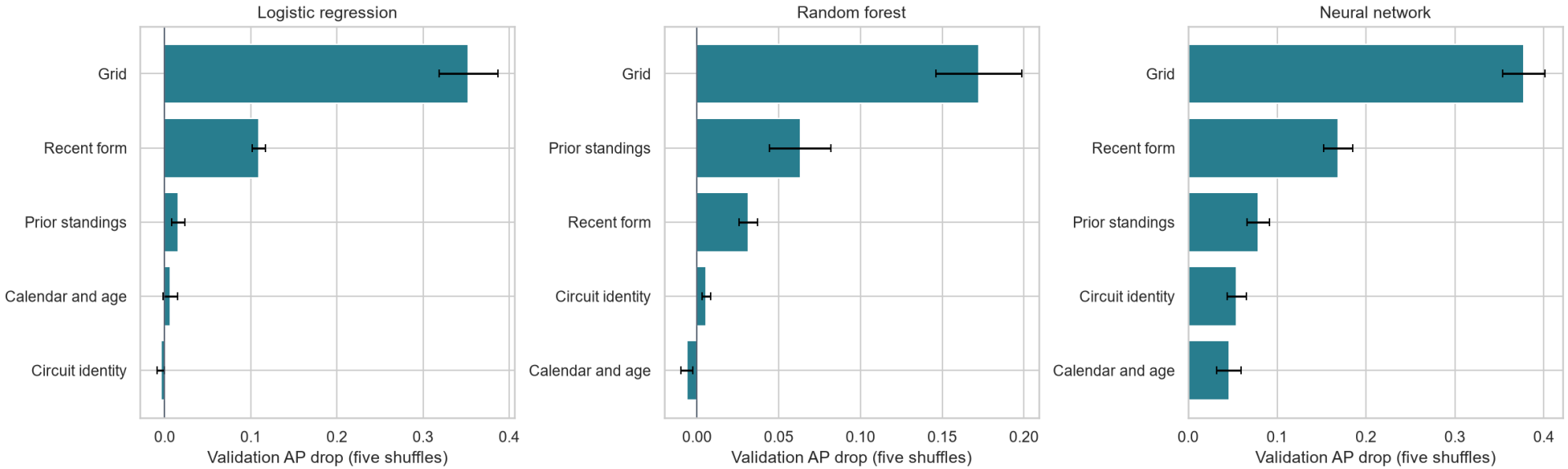
Training-row policy scores and retained coverage. Validation/test coverage is preserved for all candidates; excluded rows are characterized numerically.



Exact notebook export: 5_training_row_policies.png | Run 20261018T210548Z

Figure A44 | 6 permutation all models

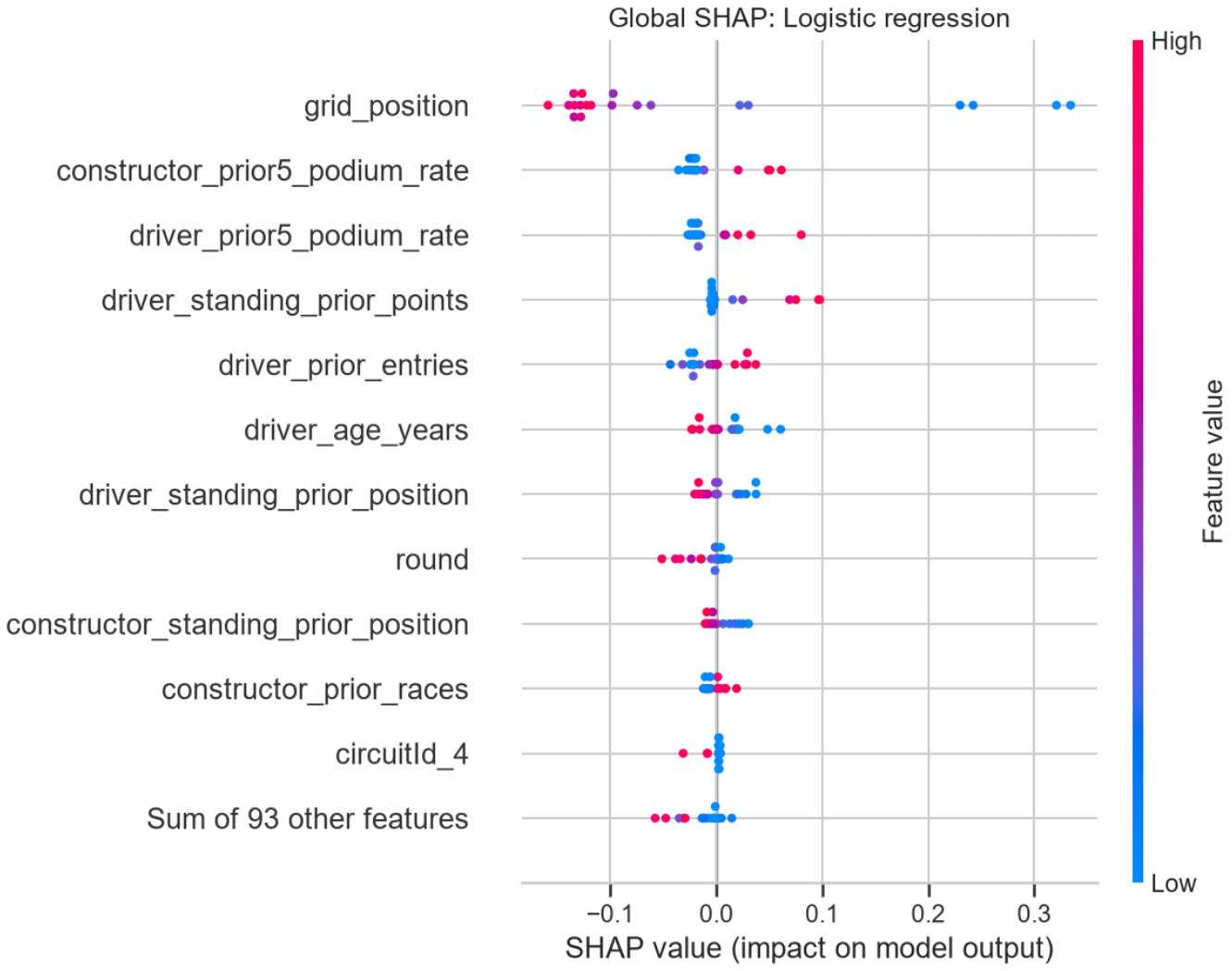
Part 6 frozen-model group permutation importance over five validation repeats. AP loss measures reliance and can be affected by correlated inputs.



Exact notebook export: 6_permutation_all_models.png | Run 20261018T210548Z

Figure A45 | 6 shap global 1

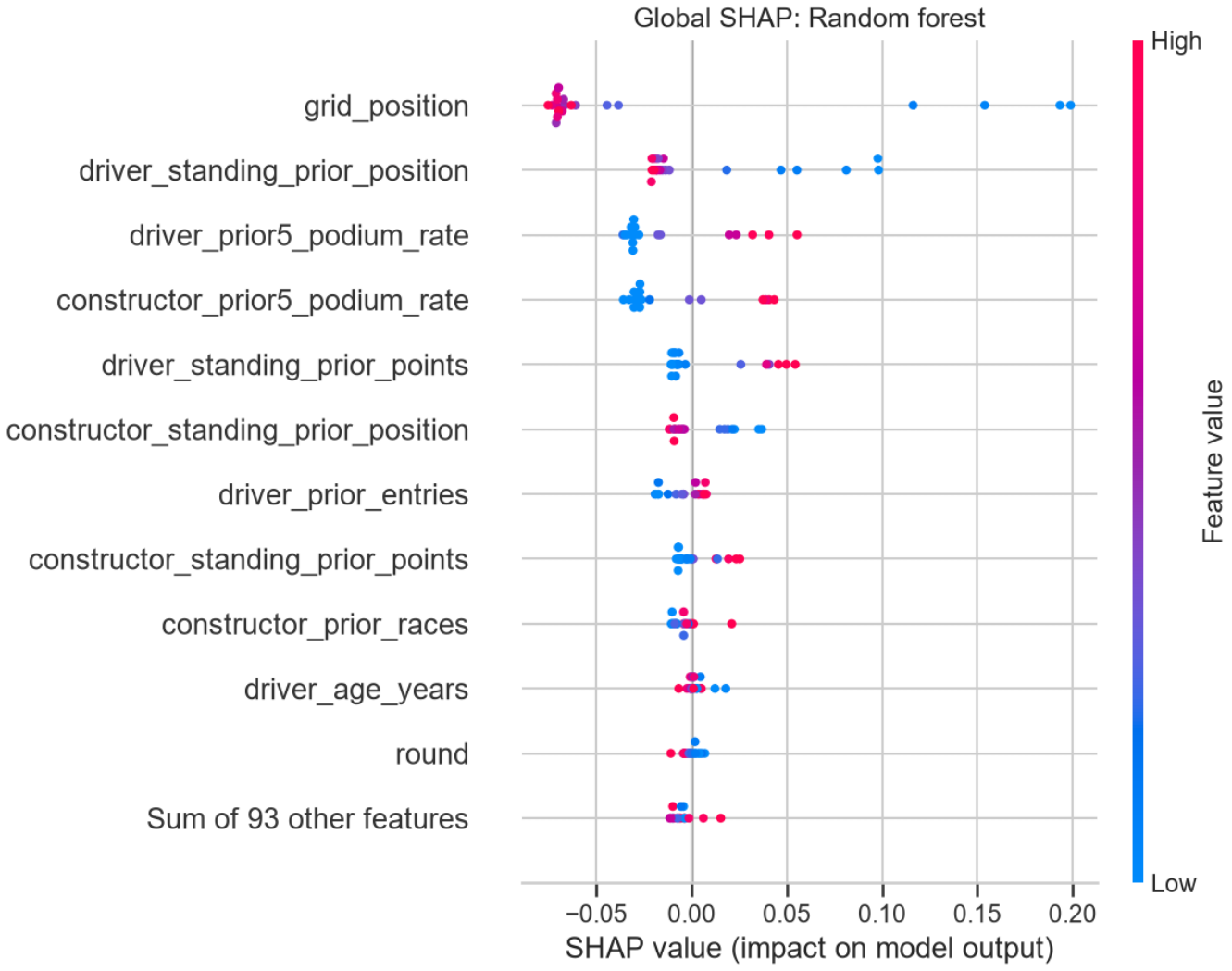
Logistic regression: global SHAP beeswarm for the common 20-row validation sample and 12-row training background. The budget is 418 evaluations per entry; effective cycles depend on invariant inputs. Prepared-column attributions are approximate and not causal.



Exact notebook export: 6_shap_global_1.png | Run 20261018T210548Z

Figure A46 | 6 shap global 2

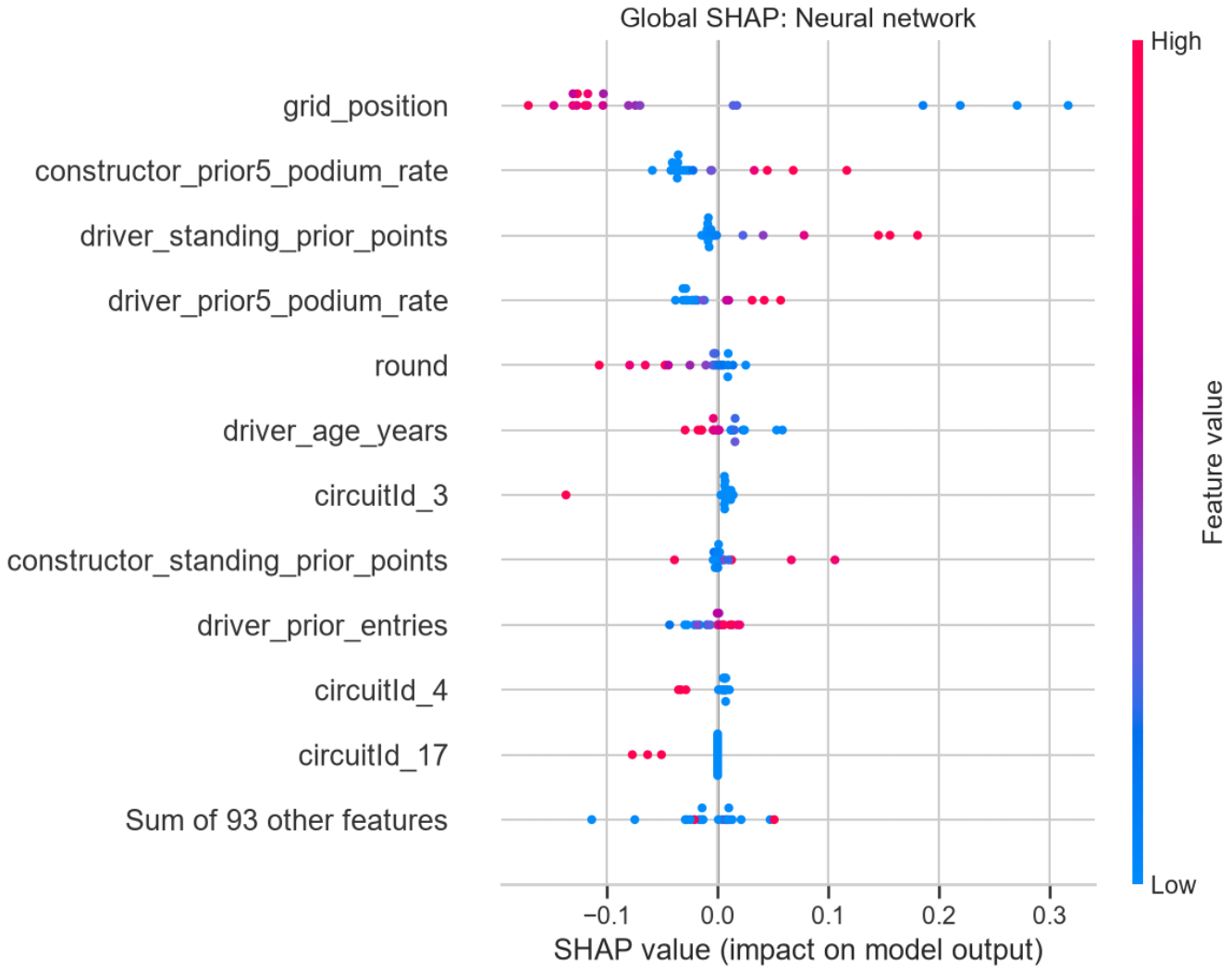
Random forest: global SHAP beeswarm for the common 20-row validation sample and 12-row training background. The budget is 418 evaluations per entry; effective cycles depend on invariant inputs. Prepared-column attributions are approximate and not causal.



Exact notebook export: 6_shap_global_2.png | Run 20261018T210548Z

Figure A47 | 6 shap global 3

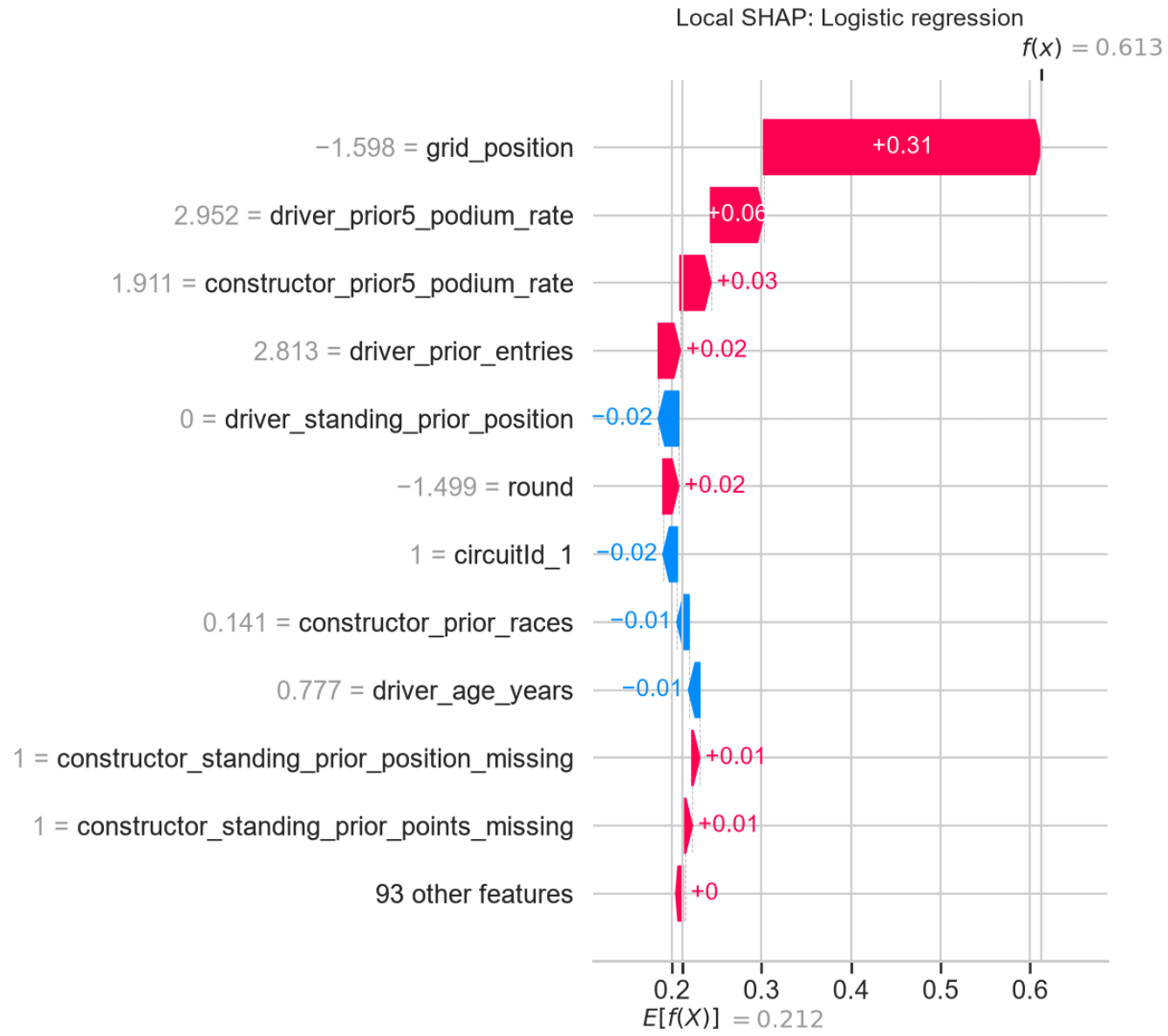
Neural network: global SHAP beeswarm for the common 20-row validation sample and 12-row training background. The budget is 418 evaluations per entry; effective cycles depend on invariant inputs. Prepared-column attributions are approximate and not causal.



Exact notebook export: 6_shap_global_3.png | Run 20261018T210548Z

Figure A48 | 6 shap local 1

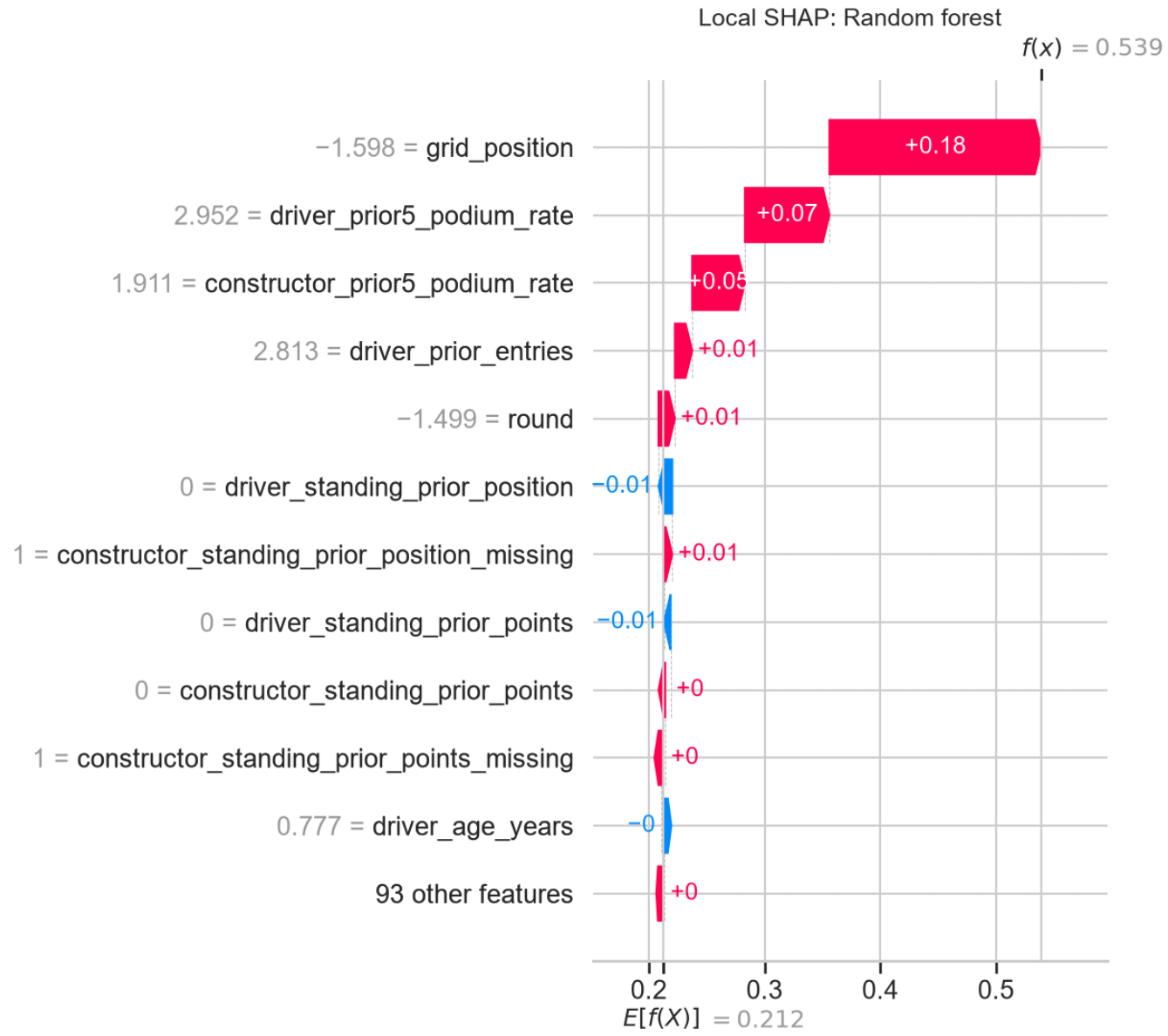
Logistic regression: local SHAP waterfall for the fixed first validation entry by sorted race/driver keys. The background plus signed contributions reconstructs the predicted probability; prepared values are not original units.



Exact notebook export: 6_shap_local_1.png | Run 20261018T210548Z

Figure A49 | 6 shap local 2

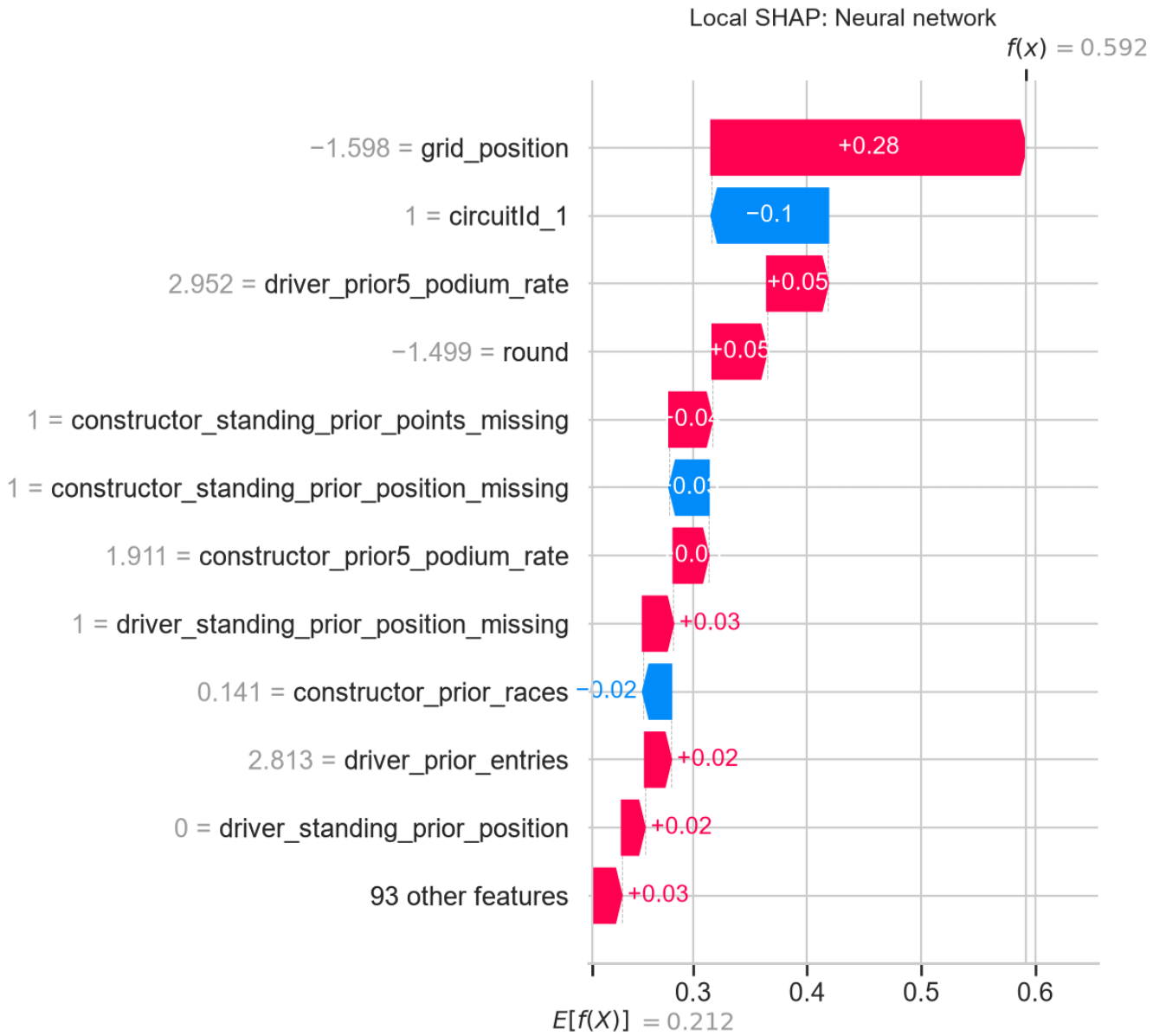
Random forest: local SHAP waterfall for the fixed first validation entry by sorted race/driver keys. The background plus signed contributions reconstructs the predicted probability; prepared values are not original units.



Exact notebook export: 6_shap_local_2.png | Run 20261018T210548Z

Figure A50 | 6 shap local 3

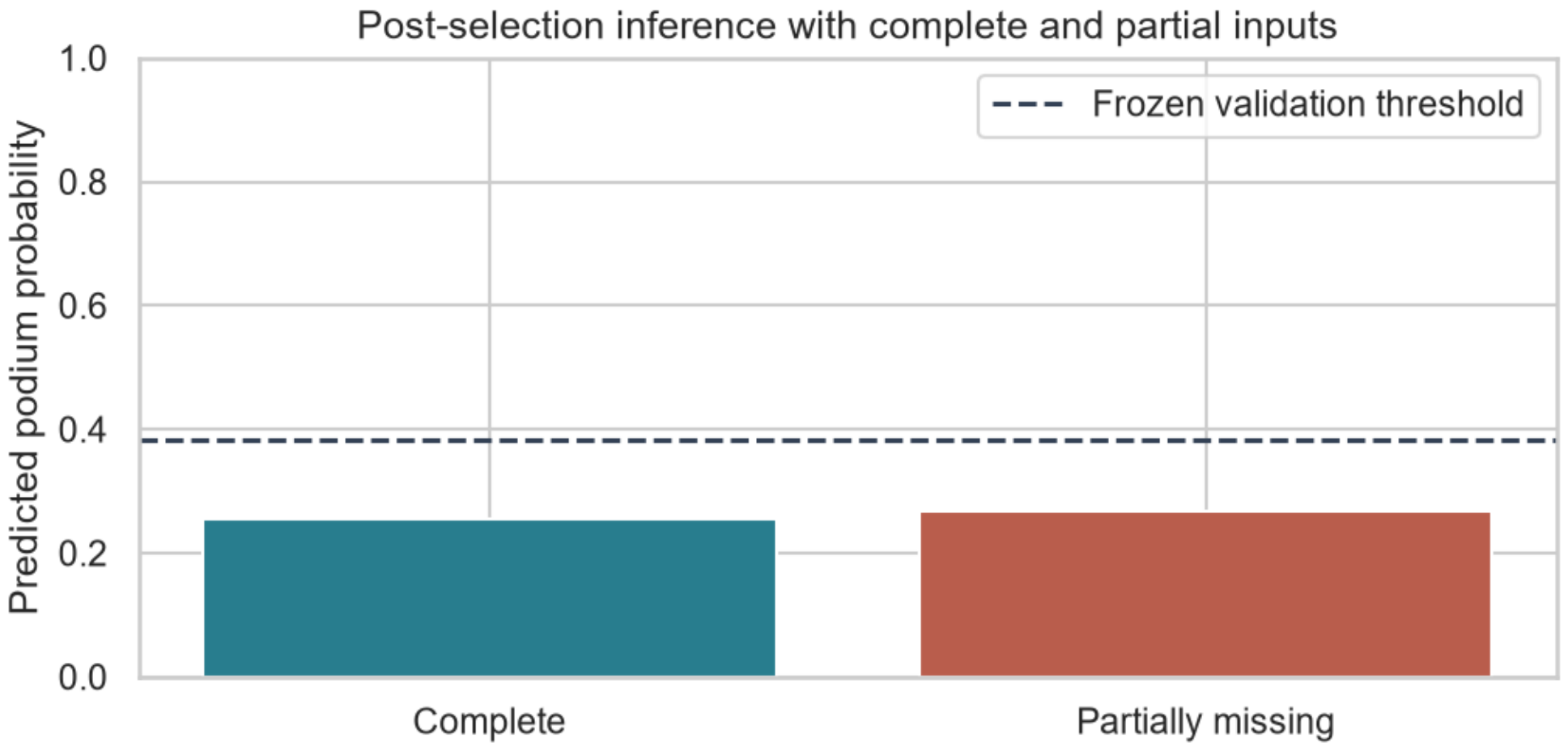
Neural network: local SHAP waterfall for the fixed first validation entry by sorted race/driver keys. The background plus signed contributions reconstructs the predicted probability; prepared values are not original units.



Exact notebook export: 6_shap_local_3.png | Run 20261018T210548Z

Figure A51 | 7 inference examples

Complete and partially missing post-selection inference examples using the frozen pipeline. Invalid inputs and unsafe dates are covered by separate boundary checks.



Exact notebook export: 7_inference_examples.png | Run 20261018T210548Z